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Learning on ecosystem services co-production in decision-making from role-playing simulation: Comparative analysis from Southeast Europe

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ABSTRACT

Role-playing simulations (RPSs)—a type of serious games—involve face-to-face mock decision making with multi-stakeholder negotiation. In decision making, they are often adopted to operate representations of central features of reality, highlighting rules and trade-offs, knowledge and learning-by-doing, motivations and negotiation processes.

This study applies an RPS to investigate the mechanisms of activation of ecosystem services (ES) flow, as a result of co-production of the actors involved in a decision-making process. Against a common setting and predetermined objectives, arrangements negotiating ES and associated motivations are analyzed in relation to the RPS characters and their roles in the ES flow. This study applies a comparative experimental approach to support a transnational capacity-building process on sustainable local development through renewable energy sources. The RPS was replicated in 10 sessions in eight Southeast European countries, with almost 230 participants from marginal rural landscapes.

From the results, we highlight the variability of the selection of ES, and the arrangements and motivations within each RPS session. ES arrangements emerge from group-based decision-making processes, where pools of players are pivotal in analyzing, assessing, and orienting decisions, beyond pre-existing expert knowledge on ES. Strengths and limitations of the method are discussed in the context of simulation practices.

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1. Introduction

Starting from the Millennium Ecosystem Assessment (MA, 2005), the ecosystem services (ES) concept has acquired growing attention in decision-making processes because it has increased the effectiveness in environmental planning and management against biodiversity loss (Hauck et al., 2013; Loft et al., 2015, Schleyer et al., 2015). The ES concept was formulated in order to link biodiversity conservation to the capacity of ecosystems to provide benefits for human well-being (Haines-Young and Postchin, 2011; Potschin-Young and Haines-Young, 2018). Moreover, the ES concept was recognized for its capacity to facilitate communication among decision makers from different sectors for multiple ES (Hauck et al., 2013), as well as in integrating economic and non-economic values of preserving ecosystems in decision-making processes (TEEB, 2010).

The cascade approach was widely adopted as the model to represent the production chain of ES and their flows (Haines-Young and Postchin, 2011), between the ES supply and ES demand (Burkhard et al., 2012; Wei et al., 2017). According to Burkhard et al. (2012), the ES supply is the capacity of a specific area to provide a set of ES in a specific moment of time because of the functions delivered by ecosystem processes and structures. Besides the different understanding given in the literature (Wei et al., 2017), the ES demand represents the ES consumption and desire in accordance with different ES categories by a human society (Wolff et al., 2015). The ES flow takes place between the demand and supply in space and time (Baró et al., 2016; Gaglio et al., 2017a, 2017b; Syrbe and Walz, 2012), and it is strongly affected by the stakeholders and actors involved in the ES delivery process (Haines-Young and Postchin, 2011; Palomo et al., 2016). The ES demand depends on the preferences, values, and perceptions of the ES beneficiaries (Martín-López et al., 2012; Wolff et al., 2015), which can be often contrasting between stakeholder groups (Duguma et al., 2011; Iniesta-Arandia et al., 2014; Lamarque et al., 2014). ES managers, also called providers (e.g. Loft et al., 2015), influence the delivery of ES with their behaviors and preferences (Gissi et al., 2014; Loft et al., 2015; Rova et al., 2017). Moreover, not all ecosystem functions are recognized as benefits (Burkhard et al., 2012; de Groot et al., 2010; Haines-Young and Potschin, 2010) and stakeholders managing an ecosystem usually benefit

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from only part of the ES provided by that ecosystem (De Groot et al., 2010). ES do not exist in isolation from people's needs (Haines-Young and Postchin, 2011; Palomo et al., 2016), but they are "contingent", that is, they depend on "particular human activities or wants" (Boyd and Banzhaf, 2007, p. 621). ES delivery emerges from the interplay and feedback loops between biophysical structures and societal needs, values, and perceptions within specific social-ecological systems. ES management can affect the supply and demand and be affected by ecosystems, socio-economic systems, and ES demand-supply mismatches (Geijzendorffer et al., 2015; Wei et al., 2017).

However, there is a general gap in understanding the links between ES functions, the resulting benefits that contribute to human well-being, and the societal responses to their provisions within decision-making processes (Loft et al., 2015). ES are vastly investigated from the side of supply but independently from demand (Wolff et al., 2015). Numerous inputs have been provided in the literature on ES mapping (e.g., Burkhard et al., 2013; Willemen et al., 2015), and understanding is growing on the side of demand (Wolff et al., 2015; Loft et al., 2015; Wei et al., 2017). Wei et al. (2017) recently reviewed 38 studies integrating ES supply and demand, to understand potential mismatches that can become the focus of ES management. They found that assessments of the demand and supply are generally performed in parallel, and then compared, as described from the methods used: biophysical indicators for ES supply, while consumption, preference, perception, or economic value indicators for ES demand (Wei et al., 2017). On the contrary, integrated research addressing the complete feedback loop between biophysical structures and biodiversity, ES flow, and decision making has remained mostly theoretical (Lamarque et al., 2014). The feedback effects from ES to human actions and the consequences on ecosystem processes are rarely taken into account (Meyfroidt, 2013). Feedbacks depend on how changes in management affect ES and how, in turn, these changes in ES are perceived by policy makers, ES managers, and stakeholders (Loft et al., 2015; Meyfroidt, 2013).

The link between the ES demand and supply is crucial particularly when dealing with ES governance, which is defined as "the institutionalization of mechanisms for collective decision-making and collective action with respect to natural resource management" (Rival and Muradian, 2013, p. 4). In order to achieve effective decisions, collaborative and deliberative approaches (Healey 1992, 1997, 2008) emphasize a discursive and interactive process between policy makers, stakeholders, and actors in a specific social-ecological system (Lennon et al., 2015; Palomo et al., 2016). Such approaches are intended as means of identifying priorities and developing strategies for collective actions around ES trade-offs. ES trade-offs are defined as alternative situations, in which the increased provision of one ES can inhibit the provision of another ES (Bennett et al., 2009; Meacham et al., 2016). The decision-making process evolves in time around the negotiation on ES trade-offs between actors and their diverse interests and value systems (Loft et al., 2015) in the process of individual and collective learning (Lennon et al., 2015). As a result, actions and measures can support some ES at the expense of others (Gissi et al., 2016). In decision making, ES result from the co-production between social and ecological systems (Jones et al., 2016; Palomo et al., 2016), meaning that the negotiation of benefits and values derived from ecosystems is influenced by the specific environmental, socio-economic, and policy settings. ES co-production is also influenced by (i) the heterogeneity of involved actors (Hauck et al., 2013; Loft et al., 2015), and (ii) available fragmented information and expertise (Jones et al., 2016; Loft et al., 2015).

This study aims to investigate the mechanisms of activation and recognition of ES from ecosystem functions involved in decision making, as a result of the process of co-production of the actors in a specific (pre-conceived) policy and environmental setting. Our goal is to demonstrate that, within decision-making processes, an emerging ES flow depends on the understanding, communication and interaction of the people involved, besides their profiles or responsibilities.

The study adopts a comparative experimental approach based on role-playing simulation (RPS), applied in 10 replicates in 8 Southeast European countries.

RPS is a type of "serious games" involving face-to-face mock decision making with multi-stakeholder negotiation (Rumore et al., 2016), where participants are engaged in solving a realistic but fabricated challenge. Serious games are commonly intended as any contest (play) among adversaries (players) operating under constraints (rules) for an objective (Abt, 1987). In decision making, RPS is often adopted to operate representations of central features of reality (Guetzkow, 1963; Rumore et al., 2016), highlighting rules and trade-offs, knowledge and learning-by-doing, and motivation and negotiation processes (Costanza et al., 2014a; Sauvé et al., 2007). Gaming approaches support the simulation of interactive problem-solving situations (Costanza et al., 2014a; Lennon et al., 2015; Rumore et al., 2016) with the dynamic participation of players in interdisciplinary settings (Crookall, 2010; Eisenack, 2013). Gaming is considered as a "heuristic method, a catalyst in transmitting information through communication" (Klabbers, 2009, p. 448) while fostering the interdisciplinary synergies necessary to effectively tackle complex social-ecological dynamics (Lennon et al., 2015), as embedded in ES flows (Costanza et al., 2014a).

The RPS applied in this study reproduces a decision-making process to support ES delivery, in order to study the ES flow that emerges from the interaction between the ES managers, beneficiaries, and decision makers (personified by characters in the RPS) with a pre-conceived biophysical context (the fabricated Canka Province) along with the ES trade-off negotiation process. During the RPS, each character (a team of players) initially constructs its individual strategy, which could entail preferences about ES. Subsequently, characters (i.e., teams) are invited to stipulate deals, such as investments and agreements, in order to achieve their targets. Deals can include ES delivery. Individual preferences are mediated and negotiated between characters while finalizing deals.

In this study, we use deals to infer players' shared priorities regarding ES. Our research goal is to place emphasis on the agreed preferences that we observe to emerge from the negotiation under equal gaming conditions and interacting mechanisms.

We explore the deals finalized in 10 replicates of the RPS in order to answer the following questions:

(1) What type of arrangements emerge from the plays? Do the types of arrangements in deals around ES depend on the stakeholders' roles (exemplified by character in the RPS), on the players' background, or on the RPS setting?
(2) Do the ES emerging from the negotiation between characters representing stakeholders in the replicates depend exclusively on the pre-conceived biophysical characteristics in the RPS? Does any relationship between ES emerge from the negotiation process?
(3) Are the ES benefits entailed in the deals uniquely directed by the (pre-conceived) RPS setting (e.g., characters' individual objectives)?

In order to answer the research questions above, we analyzed the deals emerging from the 10 replicates. In the following section, we first introduce the RPS "Buy, Sell, and Trade@TERRE" (RPST) (Section 2.1) applied in this study. Then, we explain the experiment, comprising the implementation of the RPST in game sessions

in 10 different replicates in Southeast Europe and we present the method used for analyzing the results to answer the research questions (Section 2.2). We present the results considering the arrangements supporting ES flow (Section 3.1), the ES negotiated in the game sessions and emerging patterns between ES (Section 3.2), and the ES benefits (Section 3.3). We finally discuss our findings in the context of decision-making process (Section 4.1), and we highlight some remarks on the methodology (Section 4.2).

2. Materials and methods

2.1. Role playing simulation "Buy, Sell and Trade@TERRE"

2.1.1. Conceptual model

The RPST takes place in a fictitious territory of "Canka Province" (CP), designed to support capacity building on sustainable rural development through renewable energy sources (RES). The RPST is structured around five characters representing five stakeholder groups of CP in their relationship between them and within the ES flow. In the RPST, the "characters" have individual objectives, financial resources, constraints (see Table 1), and a "trump card" intended as an exclusive piece of previous information and understanding of the context and settings (for example, local or expert knowledge), which can be shared with the other characters during the negotiation. The characters represent different types of actors (public, corporate, noprofit, and community actors) according to Sattler et al. (2016). Players are randomly grouped in a team for each character. The character's information is communicated to the players through a profile sheet for each team.

The RPST gaming mechanism is structured around the multiple benefits of preserving ecosystems of CP for the services that they can provide to the different characters, including energy from RES, with a specific focus on biomass-based energy sources (BBES), classified by the Common International Classification of ES as a provisioning service (Haines-Young and Postchin, 2011). The RPST takes into consideration the trade-offs between BBES and the provision of other ES in CP (Gissi et al., 2016, 2017). The delivery of BBES from feedstock production, generally associated with landscape homogenization because of intensive agricultural practices and monoculture (Upreti, 2004; Lindborg et al., 2009), can influence other improving economic sectors such as ecotourism, which is based on the delivery and flow of cultural ES, associated, for example, with scenic beauty of heterogeneous landscapes (Syrbe and Walz, 2012).

The gaming mechanism simulates the different roles that the characters can play in the ES flow (Table 2). Roles represent the mediated functions that the characters can potentially assume in the ES delivery (Lamarque et al., 2014; Rova et al., 2017): ES managers, ES beneficiaries, and decision makers. ES managers are "individuals or agents, such as farmers, who are in charge of supporting and managing ES provision" (Gissi et al., 2015, p. 185), while ES beneficiaries are the end users of the benefits (such as the tourist company with respect to cultural ES). Decision makers can influence the ES flow through new laws, regulations, or other policy actions. Information on benefits from ecosystems, as well as on the characteristics of the area and potential consequences of some actions is unevenly distributed between characters, to simulate the fact that (i) individuals usually do not have the complete information to make decisions (Ostrom, 2010) and (ii) they rarely have access to the same information known by others with whom they interact (Ostrom, 2011).

The characters' specific objectives (presented in Table 1) are not exclusively antagonistic or synergetic. There are potential relationships between characters (Fig. 1). The characters can decide to collaborate or not, while understanding the context settings of CP to negotiate deals according to financial resources that they dispose of, and to their role in managing the environmental resources for ES delivery. In addition to individual objectives, the characters share a common goal, which is to boost economic development and reduce poverty and unemployment, while at the same time preserving ecosystem values.

The RPST we employed is a modified version of the "Buy, Sell, Trade!" developed by WBCSD and IUCN (2008), which is a support tool for training and environmental capacity building of decision makers and local stakeholders on natural resource management in general. From the original RPS, we introduced the following modifications: (i) in the RPST setting (CP) we introduced the potential trade-off between BBES and other ES; (ii) we changed the character playing the major economic actor with Sweet Energy Development Company, which plays as major driver for BBES demand; (iii) we substitute one character with Canka Farmers and Breeders Cooperative, in order to potentially play the role of BBES producers, as well as to potentially interact with the Ecotour and Landscape Advocates because of the farming activities; (iv) we produced all the supporting materials to record the dynamics and deals emerging from the game (described in Section 2.1.3); (v) we totally changed the names of the settings and characters.

2.1.2. The simulation

The simulation refers to the mechanisms of potential interaction between the characters to elaborate deals around ES delivery. The simulation is organized around the process of interaction and negotiation of the five characters. The characters interact in a common and fixed setting of CP, with pre-established environmental structures, proprieties and related functions. According to its environment, CP is potentially able to provide some ES, before ES are used, consumed, or enjoyed by human beneficiaries (Lamarque et al., 2011). Environmental characteristics and related potential ES are described textually to the players, according to the link between structures/processes, functions, and benefits they provide (Table 3). The potential sources of ES demand are also described textually (Table 3). ES potential supply and demand are not associated specifically with any character, but referred generically to CP. The words "ecosystem services" are not mentioned, in order to avoid misinterpretations, as players are not familiar with the terms (Lamarque et al., 2011). In our study, ES are considered as "the aspects of ecosystems utilized (actively or passively) to produce human well-being" (Fisher et al., 2009, p. 645) (Table 4).

The characters are asked to interact and negotiate between each other in order to arrive at "deals" to achieve their objectives. Deals are any type of arrangement or agreement between two or more characters, which can include money exchange, as partnerships, joint ventures, payoffs, payments for ES (PES), agreements on common advocacy positions, or any other types of contracts or transactions (WCBN and IUCN, 2008). The characters are free to consider or not the inclusion of benefits derived from ecosystem structures and processes in the deals.

Deals have the purpose of supporting the achievement of characters' objectives, according to their individual motivations (Table 3). Motivations are the benefits the characters recognize to achieve with each deal. We classified motivations in order to record the elicitation process of the players in elaborating the deals with respect to the characters' objectives. Some motivations are potentially driven by the RPST setting, such as the legal and economic ones (both income/revenues and employment). Legal motivations directly reflect the response of the players to the accomplishment with new regulations in the deal elaboration (explicitly mentioned in the characters' profile). Economic motivations (in terms of income and revenues) are explicitly requested in the characters' profile, in order to maintain the running costs of activities. Economic employment benefits are explicitly mentioned

Table 1
Description of the five character profiles, with objectives, setting, constraints, and "trump cards." The profile information is communicated to the players divided in characters' teams through characters' profile sheets; each team has its own piece of information; the types of actors are categorized as public, corporate, noprofit and community actors according to Sattler et al. (2016). CH = character, CP = Canka Province, TL = traditional landscape, and Lak is the fictitious currency adopted by the RPST.

Character	Description	Objectives	Settings/Constrains	Initial Budget (Lak)	Taxes to CH1 (Lak)	Running costs (Lak)	Trump-card*
CH1. Canka Province government (public actor)	Public actor representing the local government, enforcement-driven for public interests	Overall well-being of the inhabitants of the region To maximize the economic return from local resources To face the rapid increase of energy demand in the CP Be in charge the decreasing vulnerability to natural disasters as flooding	(i) It owns the land that can be leased but not sold; (ii) Taxes from other character: 200 Lak (CH2), 100 Lak (CH3), 50 Lak (CH4);	100	–	–	It is in charge of new regulations, It collects funds from implementation of development projects
CH2. Sweet Energy Development Company (corporate actor)	Private actor for RES energy production representing private interest	To build a biomass plant (secure land) To build infrastructures and services To provide feedstock to biomass plant (secure feedstock)	(i) requires 20 km^2 in TL for the plant; (ii) comply with water regulation;	1500	200	950	It can create development because of jobs and satellite activities
CH3. Canka Farmers & Breeders Cooperative (community actor 1)	Local community actor, representing a league of small farmers acting at local scale	Looking for new stream of incomes Additional areas to cultivate to build stable demand for products Comply with water and soil regulation Maintain good relationships with authorities and local population	(i) Subjected to new environmental regulation, (ii) they own 100 km^2 in TL, 100 km^2 in IL; (iii) they must prove compliance with environmental regulation, cost for a water treatment plant is 500 Lak (to be payed to the CP government),	950	100	500	Rare species has been seen in your area of activity
CH4. Landscape Advocates (no-profit actor)	Non-governmental no-profit organization for environmental advocacy and intermediation	To lobby the government to protect, enhance and conserve healthy TL Increase overall level of environmental awareness in Canka Province Help design an innovative sustainable management plan Maintaining good relationships with all	–	200	–	50	Knowledge on ES valuation which impacts on other players: (1) revenues for recreational value of TL (from 220 to 0 Lak according to area); (2) cost for water filtration and soil prevention: 200 Lak per 200 km^2; (3) Ecol. Impact of the new plant of 20 km^2 on 100 km^2

(*continued on next page*)

Table 1 (*continued*)

Character	Description	Objectives	Settings/Constrains	Initial Budget (Lak)	Taxes to CH1 (Lak)	Running costs (Lak)	Trump-card*
CH5. EcoTour (community actor 2)	Local community actor delivering tourism services acting at local scale	To offer Ecotours in Canka Traditional landscape to bird watchers Maintaining good relationships with all	(i) requires 200 km^2 of TL to maintain the habitat for species;	150	50	80	Tourists contribute to local economy; any development can affect habitat of key species in TL; you own 180 Lak if 200 km^2 of TL are maintained

* Piece of previous information that can help to stimulate deals.

in the CP government character's profile. On the other hand, the environmental, public image, and moral and values motivations proposed to record players' elicitation process, are not explicitly mentioned in the RPST setting. Environmental motivation represents the recognition of some benefits from ecological structures or processes (i.e., ES) to achieve an individual objective. Environmental motivations, as well as ES, are not strictly related to a formal requirement of the RPST, and therefore, if mentioned, they are perceived by the players in the simulation independently from the RPST setting. Characters are asked to explicitly declare the motivations associated with deals and related ecosystem functions or ES (if any). The monetary exchange agreed in the deal represents the exchange value negotiated by the characters for the delivery of the ES supported in the deal. The simulation adopts a fictitious currency (Lak), whose value is not predetermined. As anchor point to set the monetary exchanges and prices, the players are informed about the characters' financial situation (in the profile sheet, Table 1), which is expressed through (i) the initial budget they can use to exchange deals, (ii) the costs for running the activities they are asked to maintain, (iii) the taxes to be paid at the end of the negotiation, and (iv) other expenses to be sustained by some of the characters. The characters' financial situation is reported in the character profile sheets.

2.1.3. The game play and the materials

The game play is the sequence of the gaming phases through which the simulation takes place. Participants (from now on "players") are randomly divided into five teams, i.e., one team per character. The simulation is performed in teams of an approximately equal number of players each. The simulation is organized in phases moderated by a facilitator, which coincide with the ones of the RPS from WC and IUCN (2008). Phases and materials (distributed to the players in paper sheets) are shown in Fig. 2.

After an introductory talk on the potential benefits derived from ecosystems and their functions, the chief mediator introduces briefly the game, the context setting (i.e., CP), the characters' profiles, and the rules of the game (Phase 1). Then, each team familiarizes with its character's profile, targets, and objectives, reported in paper sheets in order to produce a team internal strategy, meaning the potential actions and related deals in order to achieve their targets considering potential collaborations with the other teams (Phase 2). In Phase 3, teams are invited to meet with the other teams to discuss about potential partnerships, agreements, and actions. Players can split among different teams in order to advance the discussion with them simultaneously. There is no predetermined set of meeting pairs between teams. The discussion is set free and open according to the priorities established by each tea, and related willingness to discuss and collaborate. Each team records the meetings according to priorities in the sheet on meeting pairs. In Phase 4, each team regroups and discusses within team members about opportunities for collaboration on deals and other information of different types collected from the other teams. Phases 2 to 4 are replicated to simulate the learning process of acquiring new knowledge and information, as well as feedbacks from the other players. In our game sessions, we played 2 rounds of Phases 2–3–4. In Phase 5, team members meet to elaborate and finalize the deals they want to conclude. Deals are written in the deal sheet, where the involved teams indicate (i) the characters involved in the deal, (ii) a general description of the contents of the deal (actions, measures, etc.), (iii) the motivation/s of each character to conclude the deal (Table 3), (iv) the environmental proprieties and functions involved in the deal, if any, and (v) the monetary exchange (in Lak) involved in the deal. Finally, each team presents the agreed deals in public, in order to inform all the players about the agreements, motivations, monetary exchange, and ecosystem functions involved.

Table 2
Potential roles of characters with respect to the ES flow involved in the RPST according to the characteristics included in the characters' profiles, elaborated from Gissi et al. (2015) and Rova et al. (2017). Roles represent the mediated functions that the characters can potentially assume in the ES delivery (Lamarque et al., 2014; Rova et al., 2017). Roles are not exclusive, and they can emerge in each game play from the interactions of characters; D = decision maker, M = ES manager, B = beneficiary, and E = expert.

ES	RPGST characters				
	1. Canka Province government	2. Sweet Energy Development company	3. Canka Farmers & Breeders Cooperative	4. Landscape Advocates	5. EcoTour
1. Water purification	D		M/B	E	
2. Flood prevention	D		M/B	E	
3. Habitat services	D	M	M/B	E/B	B
4. Food production	D		M/B	E	
5. BBES	D	B	M/B	E	
6. Aesthetic values	D	M	M	E/M/B	M/B
7. Recreational values	D	M	M	E/M/B	M/B
8. Soil preservation	D		M/B	E	

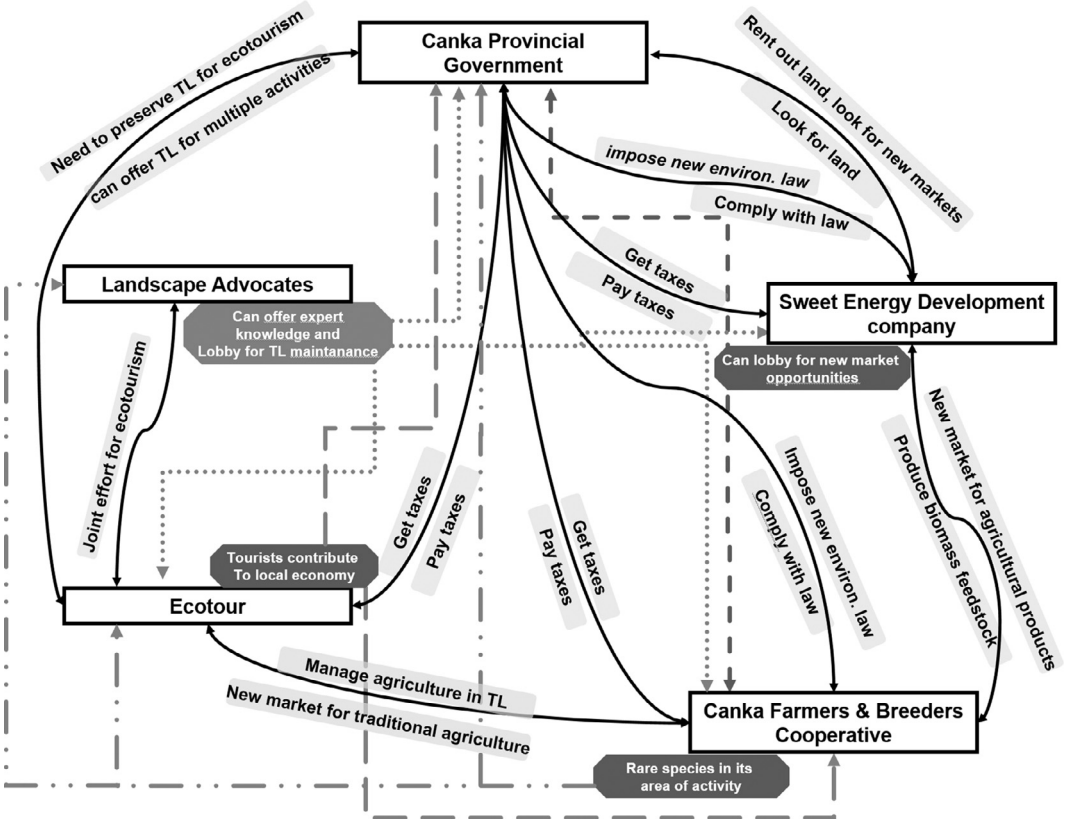


Fig. 1. Schematic of the potential relationships between characters interacting in Canka Province, which constitutes the role-playing simulation "Terre" (RPST) mechanisms. Arrows indicate potential mutual relationships between characters, and plain lines represent potential relationships because of characters' profiles, related competences, and objectives. Dashed lines are potential relationships entailed within the individual pre-existing knowledge (trump card) of each character, distinguished according to different hatchings per character; TL = traditional landscape.

At the end of the game, during the debriefing phase, the achievement of individual objectives is discussed and evaluated in the framework of the achievement of the general target of the RPST. The final discussion considers (i) if the capacity of ES delivery is conserved (no net loss), (ii) if the landscape is partially converted (net loss), such that the capacity of ES decreases due to the agreed deals, or (iii) if there is an increase in the delivery of ES because of the agreed deals.

2.2. The experiment

The RPST was played in 10 countries (Austria, AU; Bosnia Herzegovina, BH; Bulgaria, BU; Croatia, HR; Hungary case no. 1, HU1; Hungary case no. 2, HU2; Italy case no. 1, IT1; Italy case no. 2, IT2; Romania, RO; Slovenia, SL) in respectively 10 game sessions (Fig. 1). These locations were chosen because they took part in a transnational capacity-building program on sustainable local development through RES, promoted by the local institutions. The RPST was applied as a learning tool on multiple benefits emerging from ecosystems and to train local stakeholders on ES trade-offs. All the locations pertain to "rural peripheral" to "rural accessible" regions (fundamentally rural, hilly, and mountainous), according to the definition of rural area typology (Jonard et al., 2009), extended from Jonard et al. (2007) and OECD (1994). The locations are characterized by low population density and high remoteness/peripherality (the travel time by road network to

Table 3
Textual description of the ES supply capacity and potential demand in CP, which is the pre-established fabricated setting of the RPST.

ES Class*	ES	ES supply capacity potential	ES demand potential
Regulating	1. Water purification	"The traditional landscape can treat the wastewater generated by farming and breeding activities, because of the presence of aquatic vegetation dispersed in the landscape"	(1) "New national environmental regulations on wastewater discharge, nutrient cycling and soil protection was launched by the National government"
Regulating	2. Flood prevention	"The Canka landscape can potentially prevent from flooding because of the articulated mosaics of agricultural ponds and arboreal vegetation, with the presence of woods"	(1)
Supporting	3. Habitat services	(2) "A rare and endangered bird, the Red-footed Falcon (*Falco vespertinus*), nests in the Canka Traditional Landscape, and needs a significant amount (200 km^2) of it to breed properly"	(3) "LandscapeAdvocates.org has decided to push Canka Province government to make sure the traditional landscape is preserved"
Provisioning	4. Food production	"The Canka region is a marginal rural area devoted to the production of food and fiber for agricultural and bioenergy purposes."	"The Canka Farmers & Breeders Cooperative is looking for new stream of incomes, as well as additional areas to cultivate and to expand its activities"
Provisioning	5. BBES		"The Sweet Energy Development Corporation who have identified a parcel of land that is highly suitable for the construction of a biomass plant, profiting of the land availability for services and feedstock." (4) "Canka Province is keen to attract new investments and business opportunities, to differentiate the potential economic activities in its territory""There is a general increase of energy demand"
Cultural	6. Landscape aesthetic values	"Canka Traditional Landscape, a 400 km^2 large and diverse ecosystems, where traditional agriculture partly maintains the highly value landscape for citizens and tourists."	(2), (3), (4)
Cultural	7. Recreational values (rural tourism)	"Tourists from around the world come on these tours to try and catch a glimpse of this extremely rare bird, as it only lives in and around the landscape."(3)	(2), (3), (4)
Regulating	8. Soil preservation	"Traditional agricultural practices are used for soil protection and preservation, in order to maintain the natural capacity of soils to support agriculture and other functions"	(1)

* According to CICES (Haines-Young and Postchin, 2011).

urban centers). The land cover index is for the 75% assessed as "open space" or "natural" (non-artificial), namely covered by forest, agricultural, or natural areas.

The local institutions identified the players of the transnational capacity-building program between local policy makers and officers (public actors), representatives of the private sector and professionals (private actors), and the civil society (Fig. 3) through a stakeholders and target groups analysis. The players' background for each session varied significantly, both in number and in typology (Table A.1).

The RPST was initially played in English in Austria in October 2013 by the public officers of the 10 local institutions driving the transnational capacity-building program, with the authors of this study as facilitators. In this session, the public officers of the 10 local institutions familiarized with the general goals, aims, and rules of the RPST through playing. The results of this initial session were not included in the analysis. Then, the same public officers, as facilitators, replicated the RPST in their location with expanded target groups and stakeholders. The 10 simulations took place in the local languages, in order to facilitate the players' negotiation, which was conducted through verbal interaction with all the translated materials. The 10 RPST sessions involved 223 players in total.

The results of each game session were translated in English and analyzed according to the information recollected in the simulation materials, specifically considering (i) the deals sheets, (ii) meeting-pairs sheets, and (iii) participants sheets. The deals, in their textual description, were transcribed. The authors associated ES to a deal if (i) the ES were explicitly mentioned in the deal description, (ii) an ES manager was included in the deal with related managed benefit from ecosystems, and (iii) if the benefit from the ES supply-demand chain was explicitly included or mentioned. When environmental conservation was mentioned *per se,* we did not associate any ES to the deal, because conservation can support biodiversity but does not necessarily support the delivery of ES (Harrison et al., 2014).

All the information (on the type of deal, motivations, monetary exchange, and characters involved in the deal) was recorded in a unique database. Moreover, the players' background was recorded

Table 4
Description of the motivations of characters that the players must declare while negotiating a deal. The motivations are divided in two groups: (i) those that are potentially driven by the RPST setting because of the characters' objectives, and (ii) those that are not explicitly mentioned in the RPST setting, but are proposed to record the players' elicitation process.

Type of motivation	Description
Motivations potentially driven by the RPST setting	
Economic (income/revenues)	The character gets a benefit in terms of economic advantages (e.g. of income or revenues)
Legal	The character gets a benefit in complying with existing or new regulation
Economic (employment)	The character gets a benefit in terms of supporting employment for its own or for the community
Motivations proposed to record players' elicitation process, but not mentioned in the RPST setting	
Environmental	The character gets a benefit in supporting the provision of specific environmental functions, structures or processes.
Public image	The character gets a benefit in terms of its public image
Values and moral	The character gets a benefit in promoting its values and moral

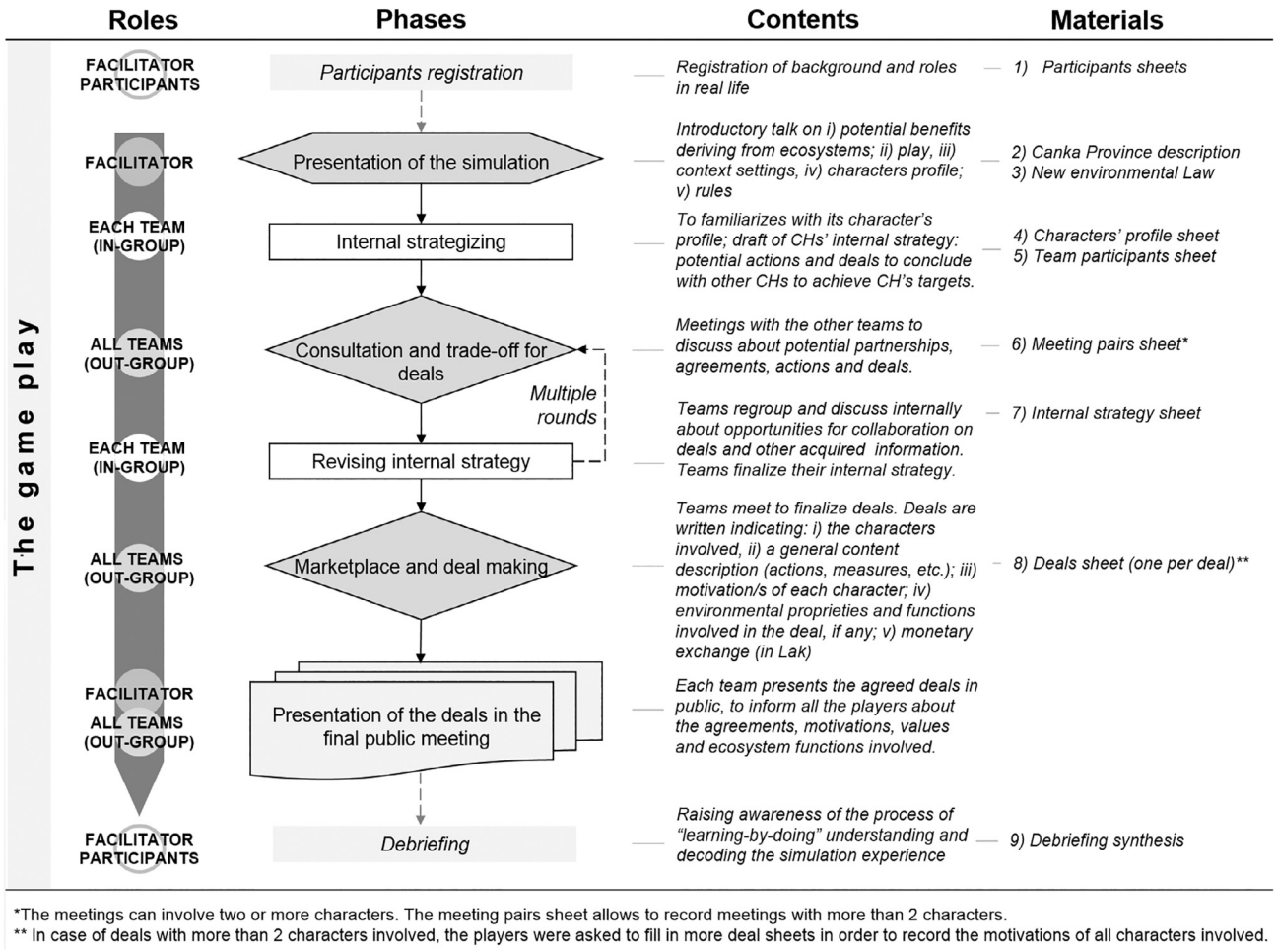


Fig. 2. Flowchart of the RPST phases, with the description of contents, roles and materials used in each phase of the simulation; CH stands for "character."

(public actors, private actors, or civil society), in order to verify the emergence of any relationship between the players' background and the type of deals produced in the game.

In order to answer the first research question, the deals were classified according to the governance arrangements categories proposed by Vatn (2010) and Sattler et al. (2016) reported in Table 5. The types of arrangements are analyzed per game session and by type of transaction. To answer the second research question, ES emerging from the negotiations were analyzed per deal and per game session, as well as the combinations of ES and related classes included in the deals. A hierarchical cluster analysis was applied to verify any emergent pattern of clustering of deals in relation to the negotiated ES. To answer the third research question, we analyzed the motivations associated with the deals in relation to the negotiated ES. Moreover, motivations were analyzed per game session, with respect to the ones potentially driven by the RPST setting, against the ones not explicitly mentioned in the RPST setting.

The deals were also analyzed statistically through SPSS (version 13.0) in order to explore if any correlation exists between all the parameters involved in the deals negotiation. Pearson's correlation was used to verify the presence of ES in the deals (dichotomous variables), and Spearman's correlation to analyze the ES in relation to the game sessions, variability of arrangement types, characters, motivations, monetary exchanges, and players' background. We finally verified the influence of each character's motivations on ES in the deals through multiple linear regressions.

3. Results

3.1. Arrangements supporting ES flow

In the 10 game sessions in the 8 European countries, the players negotiated 67 deals in total, out of which 24 do not explicitly include or support any ES (Table 6). The number of deals achieved per game session varied between 11 deals of the HR session, to 3 deals in the case of the BU session.

Deals negotiated in the game sessions concerned arrangements of very different types (the textual transcription of the deals is reported in Table B.1). Out of the 43 deals involving ES, 8 deals referred to PES, 6 to actions supporting ES provisioning, and 5 to agreements on ES provisioning or benefits exchange; 10 deals referred to the exchange of benefits related to BBES, while 4 deals referred to actions as lobbying for ES, raising awareness, and communication on ES relevance. Deals to build biomass power plants were taken in all sessions, but only in 4 sessions (BU, HR, IT1, and IT2) compensation measures were agreed in the deals. The wastewater treatment plant was built in 6 sessions (AU, HR, HU1, HU2, IT1, and IT2).

Game sessions showed a significant variability of types of arrangements (Table 6). Only the AU session presented deals of all types of arrangements, while the RO and BU sessions produced deals of only hierarchical and market-driven approaches. Deals negotiating ES were for the 62.8% of community-based approaches and related hybrids with hierarchical and market approaches (27 deals out of 43, Table 6). Five deals followed a hierarchical

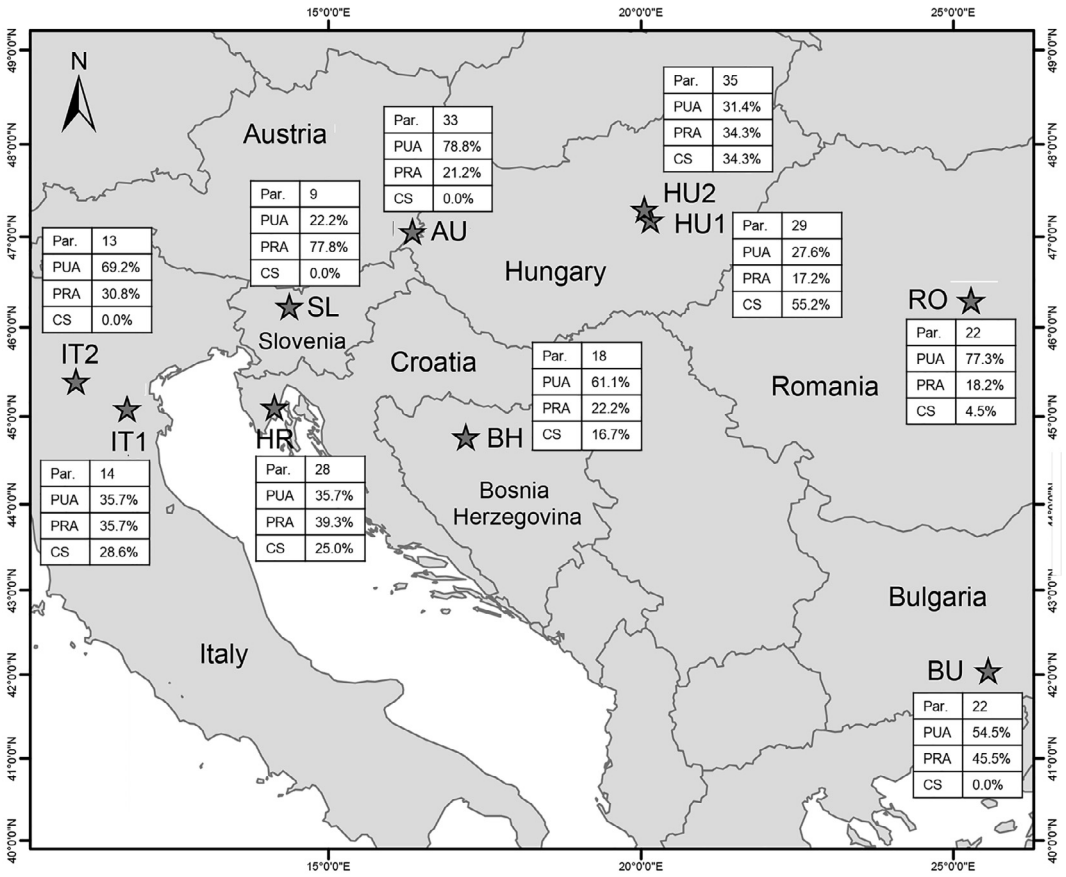


Fig. 3. Locations of the RPST game sessions in Southeast European countries: AU = Eisenstadt, Gussing district; BH = Banja Luka, Laktasi, Prnjavor, and Gradiska area; BU = Dimitrovgrad Municipality, IT1 = Rovigo Province; IT2 = Verona and its hills; HR = Cepic Polje, Istria Region; RO = Municipality of Odorheiu Secuiesc; SL = Gorenjska region (Kranj); HU1 = Municipality of Szolnok; and HU2 = Local Government of Ujszilvas. Location characteristics are fully described in Table A.1.

approach, while 11 were market-driven arrangements, mainly on BBES. The deals of hybrid arrangements of community/hierarchical and community/market approaches included 71.7% of all the ES negotiated in all the game sessions. No deal including ES was of the hybrid category of hierarchical and market-driven approaches.

The 43 deals with ES mainly involved direct monetary transactions, with payment between characters declared in 28 deals (65%, and 41% of the total number of deals) (Table B.2). Other types of rewards were tax relief (3 deals) and subsidies from the CP government (2 deals). Only one deal involved the establishment of a new tax for environmental conservation. Savings by characters for not building infrastructures, or revenues from economic activities were also mentioned in 2 deals respectively, while 7 deals concerned agreements between characters that did not involve any monetary exchange.

We run the Spearman's correlation analysis for the variability of the arrangement type for a matter of statistical significance of the sample size. The results of the Spearman's correlation analysis reported that the variability of the type of arrangements in the deals is positively and moderately correlated with the cultural ES of recreational values ($\rho = 0.414$ at 0.01 level) (Table B.3). There is also a weak correlation with the aesthetic values ($\rho = 0.373$ at 0.01 level) and weak negative correlation with BBES ($\rho = -0.374$ at 0.01 level). There is moderate positive correlation between the deal arrangement types variability and the presence of the Ecotour ($\rho = 0.401$ at 0.01 level), while a weak negative correlation emerges between the deal arrangement types variability and the RPST-driven motivations ($\rho = -0.389$ at 0.01 level); no correlation emerged with the players' background.

3.2. ES negotiated in the game sessions and emerging patterns between ES

From the negotiations in the game sessions, 43 deals mentioned at least one ES each, for 120 ES negotiated in total, and with an average of 12 ES per game session.

The ES negotiated in the 10 game sessions are shown in Fig. 4, while the ES negotiated in each deal are reported in Table B.4. Only 4 game sessions recognized all the ES classes, namely AU, IT2, HR, and HU2 sessions. On the other hand, BBES were considered in at least one deal of all sessions, followed by recreational values (9 games), water purification (8 games) and habitat services (8 games).

The same ES were negotiated several times within the same game session in different deals (Fig. 4, and Table B.5). With respect to the total number of deals, the ES that were considered and negotiated the most were the cultural ES. The recreational value ES were mentioned in 25 deals out of 67 (37.3%) in 9 games, followed by aesthetic value mentioned in 19 deals out of 67 (28.4%) in 6 games. The least negotiated ES were the food production, mentioned in 8 deals out of 67 (11.9%) in 6 games.

For the ES included in each deal, 18 deals involved the negotiation of only one ES (Table 7), out of which BBES were negotiated in 11 deals individually, followed by recreational values (3 deals), food production (2 deals), and water purification and habitat services (one deal each). Soil preservation, flood prevention and aesthetic values were never negotiated individually. The other 25 deals involved the negotiation between multiple ES, corresponding to 9 ES combinations out of which 8 combinations include at least one cultural ES (Table 7).

Table 5
Type of arrangements in terms of multilevel governance used to analyze the deals. The types are elaborated from Vatn (2010) and Sattler et al. (2016). The translation of the types in the RPST according to the RPST structure is also reported.

Type of arrangement	Definition	Examples from the RPST
Hierarchical approach	Based on systems of command in which resource allocation occurs through existing authority and power structures	Arrangements driven by the CP government, related to tax relief, subsidies, and mechanisms related to existing power structures
Market-driven approach	Voluntary exchange of goods and services among individual actors, and resource allocation occurs based on willingness to pay; main characteristics is "distribution of access to resources to trade" (seller-buyer logic)	Market-based exchange of goods and services (e.g. BBES)
Community management approach	Based on cooperation among actors, and both individual and common goals are taken into account for resource allocation, reciprocity-driven voluntary cooperation.	Arrangements between the 3 community actors (farmers, ecotour, landscape advocates), as coalitions, partnerships, etc.
Community/ Hierarchical approach (hybrid 1)	Hybrid between hierarchical and community management; it involves both government and community actors in co-management arrangements	Shared responsibilities between the CP government with the other characters for mutual benefits on individual and common goals
Community/market approach (hybrid 2)	Hybrid between market and community management; it combines community management with market-driven approaches (e.g. community-developed payments for ecosystem services)	Arrangements between all actors exchanging also market benefits (e.g. incomes/revenues)
Hierarchical/market approach (hybrid 3)	Hybrid between market and hierarchical approaches.	Private/public agreements with the exchange of market benefits (e.g. incomes/revenues)

Table 6
Distribution of deals negotiating ES per type of arrangements in terms of multilevel governance in the 10 game sessions. The types of arrangements are explained in Table 5. The total number of deals per game session is also reported.

	Number of deals negotiating ES per type of arrangements							Tot no. of deals*
	Hierarchical	Market	Community	Community/ hierarchical	Community/market	Hierarchical/market	Tot	
Per game session								
AU	1	1	1	1	1	0	5	6
BH	0	1	0	1	2	0	4	5
BU	1	1	0	0	0	0	2	3
HR	1	1	0	1	1	0	4	11
HU1	0	1	0	1	2	0	4	6
HU2	1	1	0	2	2	0	6	10
IT1	0	1	1	3	1	0	6	7
IT2	0	1	2	1	1	0	5	9
RO	1	2	0	0	0	0	3	5
SL	0	1	0	3	0	0	4	5
Tot no. of deals per type of arrangement	5	11	4	13	10	0	43	67
% of tot no. of deals per type of arrangement on the tot no. of deals exchanging ES	11.6	25.6	9.3	30.2	23.3	–	100	
Frequency of ES negotiated in deals reported per type of arrangement								
ES1 Water purification	2	0	0	8	5	0	15	–
ES2 Flood prevention	2	0	0	7	4	0	13	–
ES3 Habitat services	3	0	0	7	5	0	15	–
ES4 Food production	1	0	1	3	3	0	8	–
ES5 BBES	0	11	0	0	0	0	11	–
ES6 Aesthetic values	2	0	2	8	7	0	19	–
ES7 Recreational values	3	0	4	8	10	0	25	–
ES8 Soil preservation	2	0	0	7	5	0	14	–
Tot no. of ES exchanged per type of arrangement	17	11	7	47	39	0	120	–
% of the tot no. of ES exchanged per type of arrangement	14.2	9.2	5.8	39.2	32.5	–	100	
Average no. of ES exchanged per deal	3.4	1.0	1.8	3.6	3.9	–	2.8	

* It includes all the deals negotiated per game session, both with and without ES.

As a result of the hierarchical cluster analysis (Fig. B.1), the deals involving ES aggregate around 3 clusters, namely (i) supporting and regulating ES, (ii) cultural ES, and (iii) provisioning ES (food and BBES). The analysis closely aggregated clusters (i) and (ii). Deals related to food production and BBES are the most isolated, in line with the fact that BBES stand alone in the deals of all game sessions.

The results of the Pearson's correlation analysis reported that the ES of water purification, flood prevention, habitat, and soil preservation (supporting and regulating ES) are positively correlated (at 0.01 level) with ρ values between 0.742 to 0.957 (Table B.6). Cultural ES of aesthetic values and recreational values are correlated as well, with ρ = 0.747. Mean correlations exist between aesthetic values and habitat services (ρ = 0.615 at 0.01 level), flood prevention (ρ = 0.612 at 0.01 level), and soil preservation (ρ = 0.573 at 0.01 level).

3.3. ES benefits

Considering the 43 deals involving ES, the players reported 251 motivations, which are mainly economic (24.7%) and environmental

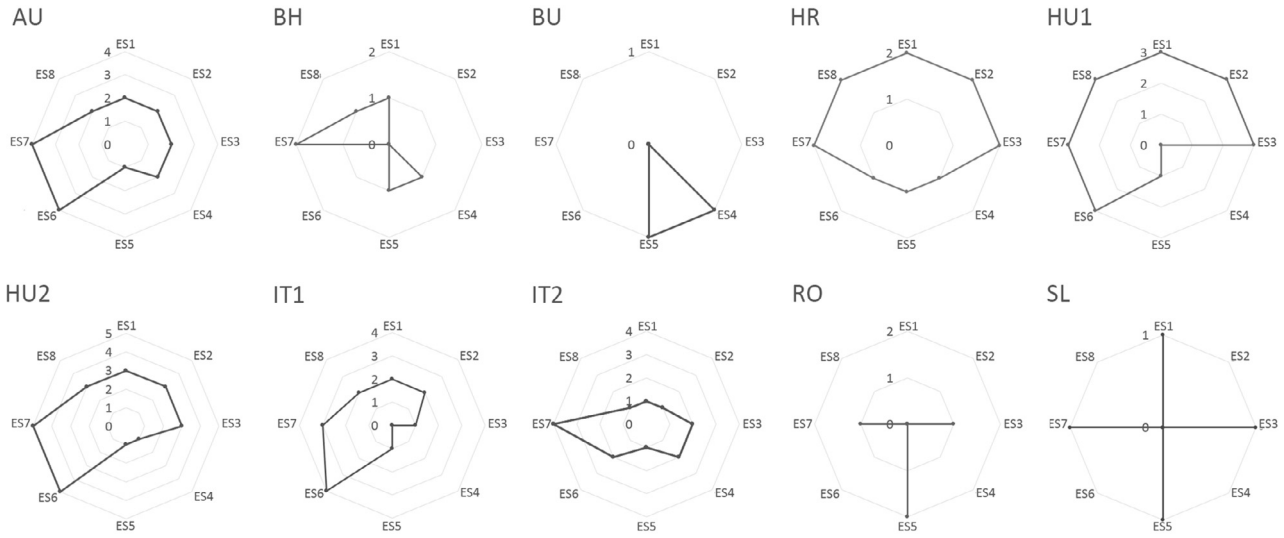


Fig. 4. ESs negotiated in the deals of each game session; when counted one, the ES were mentioned in one deal, when 2 in two deals etc. The number of deals negotiating ES and the total number of deals of each game sessions are reported; ES1 = Water purification; ES2 = Soil preservation; ES3 = Habitat services; ES4 = Food production; ES5 = BBES; ES6 = Aesthetic values; ES7 = Recreational values; ES8 = Flood prevention.

Table 7
Groups of combinations of ES exchanged in the 43 deals, and corresponding frequency.

Combinations of ES	Presence of each ES in each combination								No. of deals negotiating each ES combination
	ES1 Water purification	ES2 Flood prevention	ES3 Habitat services	ES4 Food production	ES5 BBES	ES6 Aesthetic values	ES7 Recreational values	ES8 Soil preservation	
For deals negotiating one ES									
(a) only ES1	x	–	–	–	–	–	–	–	1
(b) only ES3	–	–	x	–	–	–	–	–	1
(c) only ES4	–	–	–	x	–	–	–	–	2
(d) only ES5	–	–	–	–	x	–	–	–	11
(e) only ES7	–	–	–	–	–	–	x	–	3
Tot.									18
For deals negotiating 2 or more ESs									
(f) with 2 ESs	–	–	x	–	–	–	x	–	1
(g) with 2 ESs	–	–	–	x	–	–	x	–	2
(h) with 2 ESs	–	–	–	–	–	x	x	–	7
(i) with 3 ESs	x	–	–	–	–	–	x	x	1
(l) with 3 ESs	x	x	–	–	–	–	–	x	1
(m) with 4 ESs	–	–	x	x	–	x	x	–	1
(n) with 5 ESs	x	x	x	–	–	x	–	x	2
(o) with 6 ESs	x	x	x	–	–	x	x	x	7
(p) with 7 ESs	x	x	x	x	–	x	x	x	3
Tot.									25
Total no. of deals negotiating ES									43

ones (22.7%). Legal motivations are generally the least represented in the deals (9.2% of the sample), and generally the least mentioned per ES (mean value of 6.7%) (Table 8). Each deal involving ES is mostly supported by the co-presence of environmental (mean value of 27.5%) and economic (mean value of 25.4%) motivations, except for BBES, which are mainly motivated economically (33.8%) and because of employment (24.6%). The highest number of motivations is reported for deals involving cultural ES of recreational (151, 60.2%) and aesthetic values (107, 42.6%).

The motivations reported per game sessions are unevenly distributed (Fig. 5, Table B.7). In the case of RO, BU, and HU1 sessions, the motivations are dominated by the economic, legal, and employment ones, potentially driven by the RPST setting. In the case of HR, SL, AU, HU2, and IT1 sessions, the motivations are dominated by environmental, public image, and values and moral ones.

We verified the influence of the characters in the motivations (Table. B.8). The Landscape Advocates showed moderate positive correlation with environmental motivations ($\rho = 0.499$ at 0.01 level). The Farmers and Breeders Cooperative and the Ecotour did not show any correlation with the motivations. The other characters showed weak correlations with some motivations: CP Government with legal motivations ($\rho = 0.320$ at 0.01 level), while Sweet Energy Development Company (CH2) had economic motivations ($\rho = 0.310$ at 0.01 level).

We also verified the influence of each character's motivations on ES in the deals through multiple linear regressions. We repeated the regression considering each ES as a dependent variable of the 35 independent variables representing the characters' individual motivations. The analysis resulted in 8 significant models (Table B.9) influenced by the economic motivation of Sweet Energy

Table 8
Motivations mentioned in the deals involving the different ES classes. The motivations are grouped into those that are potentially driven by the RPST setting, and those that are not explicitly mentioned in the RPST setting. Motivation frequency per ES class is also reported.

	Motivations potentially driven by the RPST setting			Motivations not explicitly mentioned in the RPST setting				Motivations' Frequency per ES class
ES class	M1 economic %	M2 legal %	M3 employment %	M4 environmental %	M5 public image %	M6 values & moral %	Tot %	no.
1. Water purification	22.7	8.0	12.5	28.4	12.5	15.9	100.0	88
2. Flood prevention	24.6	4.3	14.5	30.4	10.1	15.9	100.0	69
3. Habitat services	25.0	5.4	10.9	30.4	13.0	15.2	100.0	92
4. Food production	23.9	6.5	19.6	26.1	10.9	13.0	100.0	46
5. BBES	33.8	10.8	24.6	13.8	10.8	6.2	100.0	65
6. Aesthetic values	26.2	5.6	14.0	31.8	9.3	13.1	100.0	107
7. Recreational values	23.8	6.6	12.6	29.8	13.2	13.9	100.0	151
8. Soil preservation	22.8	6.3	13.9	29.1	11.4	16.5	100.0	79
Average	25.4	6.7	15.3	27.5	11.4	13.7		

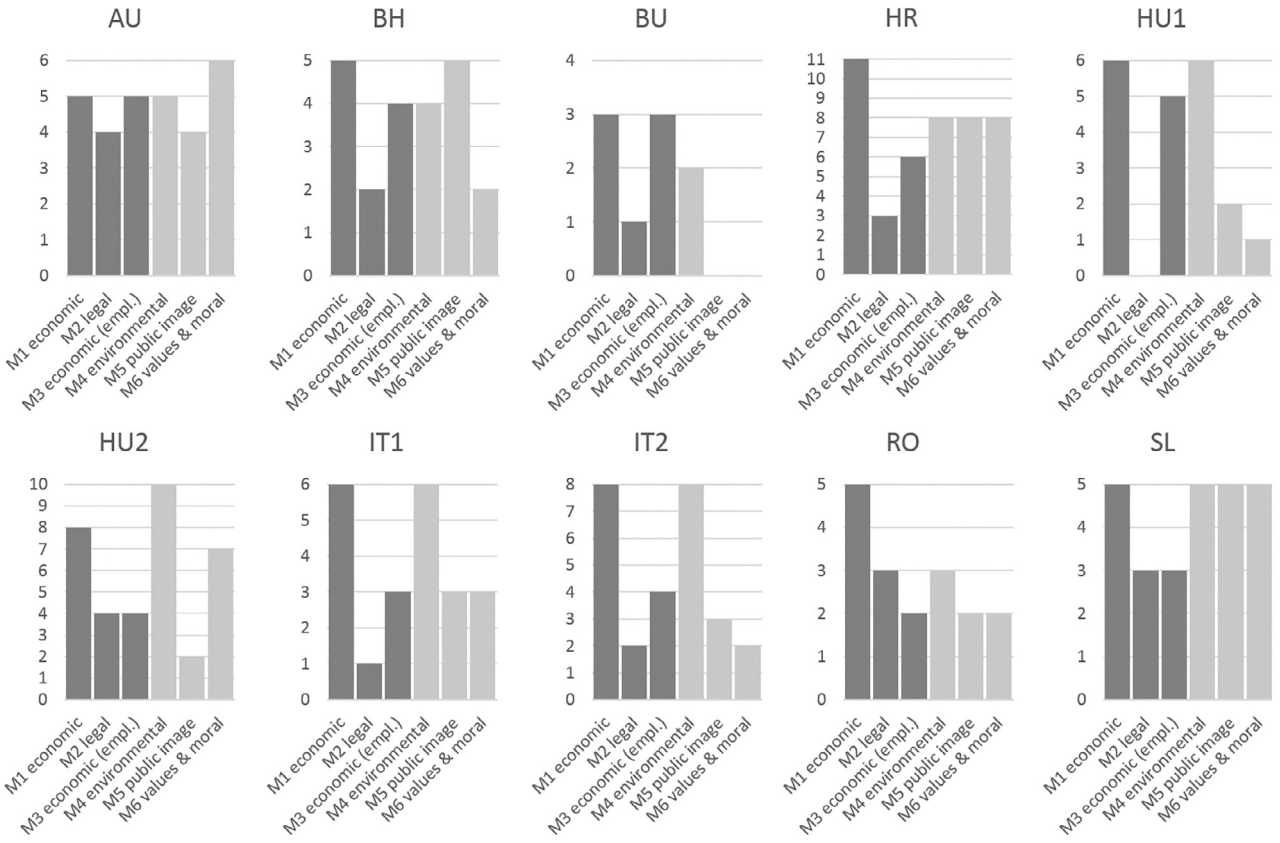


Fig. 5. Motivation distribution in each game session. The motivations potentially driven by the RPST setting, mentioned in the characters' profile objectives are indicated in dark gray, while the motivations not explicitly mentioned in the RPST setting are indicated in light gray. M = motivation; M1 = economic; M2 = legal; M3 = employment; M4 = environmental; M5 = public image; M6 = values and moral.

Development Company (CH2), which ranged from the 14.5% for ES4 (food provisioning), to 43.7% for ES6 and ES7 (cultural ES). Only BBES are directly influenced (beta = 0.546), while all the other ES are inversely influenced, with the highest beta value for cultural ES (beta = −0.661).

With respect to the monetary exchanges, the amounts of Laks exchanged in the deals are significantly variable between deals and plays, as they are associated with the different types of arrangements, ES combinations and related transactions (Table B.10). The only ES for which the monetary exchange can be comparable are the BBES, because they were always exchanged in isolation. Their monetary exchange ranges between 100 Lak and 500 Lak in the 11 deals from the 10 games.

4. Discussion

4.1. Negotiating ES arrangements in decision-making process

This study analyzed the formulation of arrangements for the ES flow within a decision-making process in a pre-conceived setting

through a role-playing simulation. The RPST represents a fabricated reality that mocks the potential mechanisms that could take place in real life. We focused on the analysis of the deals emerging from the interaction between characters—personified by teams of players—negotiated in order to achieve their objectives.

The types of arrangements reported in the deals seem not to depend on the RPST setting, because they varied significantly between the game sessions. In our experiment, multiple forms of governance emerged from the different sessions. In fact, the AU session presented deals of all types of arrangements, while the RO and BU sessions produced deals of only hierarchical and market-driven approaches. The other 7 sessions produced a mixed variety of arrangements between hierarchical, market-driven, and community-based approaches. Such variety came up from the interplay of characters starting from the same RPST with the same pre-established objectives, while interacting in the same pre-conceived and fabricated setting. The RPST was designed to provide previous knowledge on the potential supply and demand of ES in the CP. It has been already demonstrated that governance usually takes place with different combinations of hierarchical, market-based, and community-based approaches (Sattler et al., 2016). However, the ES flow also turned to be very different in the different sessions. We discovered that in the decision-making process, represented by the negotiation of ES deals, the ES delivered and the type of associated arrangements varied significantly between game sessions. In our experiment, the variability of deals and ES entailed in the deals represents a group-based, rather than individual-based variability. In fact, the deals and related ES were not elicited by individuals (players) or appointed by characters (personified by teams of players) alone. Players' interactions generated novel insights and solutions for the delivered ES, based on a new shared understanding of trade-offs and synergies between ES.

Unexpectedly, we found that the types of arrangements in the deals were correlated to the presence of the Ecotour character. In the RPST, two characters were meant to potentially influence the types of deals: (1) the CP government, as the public actor concerned with public interests, both environmental and economic ones, and (2) the Landscape Advocates character, described as a nongovernmental noprofit organization for environmental advocacy and intermediation, with the specific objective of helping in designing innovative solutions for environmental awareness and conservation. The fact that the types of arrangement are influenced by the Ecotour, instead of the CP Government and the Landscape Advocates, suggests that the arrangements were not driven by the RPST setting. It has been stated that actors from different societal spheres (civil society, market or the government) are predestined to provide different things and perform different roles in governance setting (Strachwitz, 2010 in Sattler et al., 2016), with some actors contributing mostly according to their competences and responsibilities. On the contrary, in our study, arrangements on ES delivery were influenced by the local economic actor at community level, the Ecotour. This finding seems to suggest that other actors can influence the elaboration of arrangements on ES delivery, besides politicians or prominent NGOs as suggested by Swallow et al. (2010), Rosenthal et al. (2015), Ruckelshaus et al. (2015), Waite et al. (2015). The contribution of the actors can emerge variously along with the negotiation process. The importance of the stakeholders' engagement in the decision-making process not only resides in the enhancement of the legitimacy or buy-in of the arrangements (Rosenthal et al., 2015; Ruckelshaus et al., 2015; Waite et al., 2015). Engaging stakeholder in decision-making process is useful also to uncover the unknown and unexpected contributions of the actors, which not only depend on the pre-existing government and market-based structures.

With respect to the ES emerging from the negotiation between characters, they do not exclusively depend on the pre-conceived biophysical characteristics in the RPST. In the CP, the potential capacity to provide ES was predefined with the setting of the RPST (Table 3). The textual description of the ES supply was provided to the players equally in all game sessions, through textual description representing the potential relationship between ecosystem structures and processes, functions, and related benefits. We would expect that, from the same RPST setting, a potentially similar group of ES would have emerged in the different sessions. However, the ES negotiated in each session varied significantly (Fig. 4), both in number of ES and in classes per session. Considering, for example, the two sessions that negotiated the same number of deals, IT1 and HU2 sessions negotiated respectively 15 ES and 24 ES from different classes in 6 deals (Table 6, Fig. 4). The basic scientific understanding of the ecological processes and interactions among ES is addressed as a key enabling factor to set ES arrangements as payment for ecosystem services (Huber-Stearns et al., 2017). Although the demand and supply of ES is the same per session (provided by the RPST), our experiment demonstrates that the negotiation of ES between characters involves a combination of arrangements and ES, which are different in each session; this is independent of the previous knowledge on the ES demand and supply provided by the RPST. This could be probably because the players exchanged views, considered evidence, reflected, negotiated and persuaded (Lennon et al., 2015) while interacting to achieve their objectives. Moreover, as information was scattered between characters, it is likely that the expert knowledge did not pass (or was not perceived as relevant) from the players. In any case, great effort is given in the literature to the expert-driven assessment of the ES supply (e.g., Maes et al., 2016) and demand (e.g., Wolff et al., 2015) in the scoping phase (Huber-Stearns et al., 2017) often independently (Wolff et al., 2015; Wei et al., 2017). Our findings suggest that the expert knowledge on the potential ES demand and supply was somehow influenced by the dynamics taking place in the simulation because of the interpersonal negotiation mechanisms, the understanding of the setting by the players and the preferences and perceptions the players succeeded in implementing in the interplay with the others. In decision-making practice, much more attention should be devoted in the mechanisms of negotiating ES perceptions and preferences between actors, and the relative uptake of scientific and expert knowledge derived from ES assessment in real decision-making processes.

In addition, we found some relationships between ES, which seem to emerge from the results of the negotiation processes simulated with the RPST. First, we found that cultural ES were negotiated the most, potentially because they were easily understood by the team players, as already suggested by Plieninger et al. (2013). On the other hand, BBES were negotiated mainly alone in market-driven arrangements in every session. These specific market-driven arrangements saw the major economic actor, the Sweet Energy Development Company, buying feedstock from the Canka Farmers and Breeders Cooperative, the local community actor representing a league of small farmers. BBES emerged as a rival and excludable good, and this made it easier to design PES, as suggested by Huber-Stearns et al. (2017). On the contrary, the 32 deals not negotiating BBES reported various arrangements of different types, involving different combinations of ES, which were mainly cultural, supporting and regulating ES classes (as reported in Table 6). The deals characterized by hybrid community-based approaches mixed with market or hierarchical ones reported a higher number of ES negotiated together (with an average of 3.6 and 3.9 ES respectively, Table 6), involving all actors, which were meant to support the delivery of multiple ES. Defining ES as economic goods or services was appointed as an enabling condition for the PES (Sattler and Matzdorf, 2013; Huber-Stearns et al., 2017). However, our results suggest that this may direct potential

arrangements only to maximize one benefit, in our case the BBES, toward the market-driven benefit intended in the logic of "seller-buyer" (Vatn, 2010). In our experiment, this emerged particularly for the ES that had an economically powerful direct buyer (as the BBES with the Sweet Energy Development Company). In fact, the other provisioning ES, the food provision, for which there was no direct buyer, was only exchanged individually in 2 deals out of 8, and more often exchanged with other ES.

Moreover, we found that the negotiation of ES took place around combinations of ES, giving place to arrangements involving multiple ES delivery. Excluding the 11 deals on BBES, 25 deals out of 43 negotiated multiple benefits derived from ecosystem structures and processes at once, namely multiple ES. Studies on ES supply and demand usually focus on single ES, which are mapped and assessed individually (Wei et al., 2017). Only recently, much more emphasis has been given on studying trade-offs and synergies between ES (Bennett et al., 2009; Gissi et al., 2016, 2017; Howe et al., 2014; Raudsepp-Hearne et al., 2010; Spake et al., 2017). Our findings suggest that in decision-making processes, ES need to be assessed in bundles as they are interdependent because they are negotiated in combination, besides their biophysical or social interdependences (Gissi et al., 2017; Spake et al., 2017). Regulating services—water purification, flood prevention, habitat and soil preservation—were usually negotiated together, as well as cultural ES, namely aesthetic and recreational ones. Interestingly, regulating ES were often associated with cultural ES rather than with provisioning ES. This might suggest that matching multiple values from different actors can not only support the achievement of PES (Muradian et al., 2013; Sommerville et al., 2009), but also support the delivery of multiple ES at once. Moreover, because regulating ES are correlated with cultural ES, a potential driver for achieving ES arrangements for multiple ES would be to leverage regulating ES in relation to cultural ES. While BBES and food provisioning have a market to establish their exchanged direct values, known as marketed ES (Costanza et al., 2014b; Kubiszewski et al., 2017), the two ES classes, i.e., regulating and cultural ES, are the ones that are not associated directly with market values (Daly, 1998) and activated multiple arrangements of ES. Our results confirm that cultural and regulating ES are usually associated in landscape conservation against provisioning ES (Maes et al., 2012; Raudsepp-Hearne et al., 2010).

With respect to the motivations supporting ES, we found that the ES benefits were not uniquely directed by the pre-conceived RPST, namely the characters' individual objectives. Looking at the motivations per game session, we found that the motivations potentially driven by the RPST setting were mixed with the motivations not explicitly mentioned in the RPST setting. This means that other sets of values may have entered into the negotiation (Panoutsou, 2008; Arias-Arévalo et al., 2017), most likely brought in the negotiation by the players as real-life stakeholders with proper understanding and perceptions. This finding confirms the importance of considering the multiplicity of values related to ES flow (Gómez-Baggethun and Barton, 2013).

On the other hand, with respect to the influence of the characters' individual objectives, as expected, legal motivations were correlated to the presence in the deals of the CP government; economic motivations were correlated to the major private corporate actor, the Sweet Energy Development Company, and the environmental ones to the NGO character, the Landscape Advocates. Unexpectedly, the motivations were not correlated with the Farmers and Breeders Cooperative and the Ecotour characters, the local community actors with private interests, as well as environmental ones. On the contrary, the economic motivation of the Sweet Energy Development Company influenced BBES directly and all the other ES indirectly. In the ES flow, the three characters (Sweet Energy, Ecotour and Farmers) represented the managers of ES, as well as potential beneficiaries of some ES. Our results suggest that the mix of the economic and environmental interests and power position of the ES managers can be decisive when negotiating ES. Different actors, economic sectors and policy fields can put forward multiple rationales for different ways of managing ES (Loft et al., 2015). However, we found that the most flexible actors in building such rationales were the community-level characters with private economic and environmental interests but with less power, namely the Ecotour and the Farmers and Breeders Cooperative. In fact, they were more open in negotiating different arrangements to achieve their objectives. The engagement of ES managers in the mechanism of negotiation for ES delivery is not only essential for incorporating their preferences and values (Kaye-Zwiebel and King, 2014; Panoutsou, 2008; Wei et al., 2017; Wolff et al., 2015) or for the legitimacy and buy-in of arrangements (Rosenthal et al., 2015; Ruckelshaus et al., 2015; Waite et al., 2015). Stakeholders' engagement is essential because it allows ES managers to build their motivations within the decision-making process in order to enlarge the pool of potential solutions for ES delivery while incorporating multiple benefits and values emerging from the players.

Finally, the new national environmental legislation on wastewater discharge, nutrient cycling and soil protection, which we designed in the RPST as the main driver for the delivery of regulating ES, was not perceived as such by the team players, unexpectedly. Our results show that the legal motivations were the least mentioned in general, in the deals negotiating ES. Strong institutions (e.g., laws, policies, and norms) can help the enforcement and adjudication of arrangements for the delivery of ES (Huber-Stearns et al., 2017), ensuring conditionality and providing clear and accepted mechanisms for achieving PES schemes (Muradian et al., 2013; Sattler and Matzdorf, 2013; Ruckelshaus et al., 2015; Waite et al., 2015). However, we found that the establishment of new laws was not perceived by the players as a motivating factor for stipulating arrangements on ES. Nevertheless, our results suggest that motivations for stipulating ES arrangements were more likely to be found in the systems of norms and values of the players, rather than in the pre-established objectives of the characters or external drivers (as norms or new laws).

4.2. Remarks on the methodology

In our study, we used a role playing simulation (RPS) to investigate on ES governance arrangements. The advantage resides in the fact that RPST reproduces a decision-making process around ES trade-offs between characters, personified by teams of players. Role-Playing Games (RPG) and Simulation Games (SG) are usually intended to simulate the interactions between characters in social-ecological systems in time (e.g. Villamor and Badmos, 2015). The simulation mechanism implies updating the context by taking into account the consequences of the deals made during a first round by the players, and then letting the players enter into a second round of discussions/deals-making while facing an updated situation. Our study followed other examples in the literature, called "Role-Play Simulations" (RPS) (e.g., Rumore et al., 2016), where the players cannot see the implications of their choices in a dynamic manner in time, and the game ontology only reproduces the interplay between real-life characters. On the other hand, we did introduce the time dimension entailed in the decision-making process. In fact, we simulated the process of acquiring new knowledge and understanding from other characters' preferences along with the multiple rounds of consultations between characters to elaborate deals.

The simulation was meant to reproduce the decision-making process as a collective learning process (Lennon et al., 2015; Rival and Muradian 2013), where players shared views and

information associated with the characters. RPSs show several advantages in activating learning-by-doing processes, which is essential in order to engage stakeholders in setting and implementing ES governance (Rosenthal et al., 2015). At the end of the RPST, the players were guided to become aware of the process in which they were engaged while decoding the simulation experience. The self-reflection on the game play and on the interactions between players reveals not only the game ontology, but also makes them aware of the mechanisms of negotiation between players. The process of negotiation for achieving arrangements around ES is central for setting ES governance in relation to the ES flow and all the actors involved in the provision of ES.

Another advantage for setting ES governance resides in the fact that the RPST is based on direct personal interaction between players, which is essential to support capacity building for applied knowledge as in personal and communication skills (Lennon et al., 2015). Social learning, technical, or organizational innovation are considered as expected outcomes of RPGs (Costanza et al., 2014a; Davies et al., 2015; Souchère et al., 2010; Voinov and Bousquet, 2010), while computer-assisted games may inhibit the benefits of direct personal interplay, as players usually act with the mediation of computers (Le Marc, 2010). In our experiment, we found that the information on ES supply and demand, scattered between characters, was not used in all the game sessions. For example, compensation actions for the biomass-based energy power plant were only mentioned in deals of 4 sessions out of 10. This implies that, in decision-making processes, expert scientific knowledge requires specific skills of the actors involved in the negotiation, in order to be activated. Soft skills are defined as interpersonal, human, people, or behavioral skills necessary for applying technical skills and knowledge in a management process (Muzio et al., 2007; Rainsbury et al., 2002). RPS can be used as a suitable tool to support stakeholders' capacity building in the understanding of the importance of soft skills within negotiation processes.

On the other hand, RPSs are used to address resource management challenges (Davies et al., 2015) and to explore local knowledge and human behavior, which are not easily captured using conventional survey techniques (Costanza et al., 2014a; Voinov et al., 2016), while soliciting decision rules through direct observation of the player's behavior (An, 2012). However, the use of RPSs in real ES governance setting should be carefully evaluated. The simulated environment, representing reality, can bring the risk of missing the integration of important social values into the play (Davies et al., 2015). Our experiment was not intended to represent a real setting and exploring related decisions for real life, that is "to represent real complex systems with soft tools such as RPG" (Villamor and Badmos, 2015). Our goal was to simulate the co-production mechanisms of ES flow between actors in a pre-conceived setting. We suggest the need for integrating the dialogue between actors involved in managing the ES supply with the ones expressing a potential ES demand when elaborating arrangements for ES governance. Our results suggest that, within decision-making processes, the ES flow emerges from the interplay of actors, and cannot be formulated or predicted only by matching studies on the ES supply capacity or on the ES demand potential independently. The RPST provides the opportunity to test an interactive problem-solving approach for dynamic participation in interdisciplinary settings (Crookall, 2010; Eisenack, 2013; Lennon et al., 2015) as the proper context to set a decision-making process to support ES delivery.

The methodology has some limits with respect to representing the real-life ES governance framework. We could probably arrive at patterns of ES relationships if we reproduce the simulations in more cases within the selected countries. On the other hand, we have not found any relevant similarity between games repeated in Italy and in Hungary in two different case study areas. However, the number of game sessions does not inhibit the general goal of the study, as we analyzed the deals in relation to the total number of players and characters' roles. In the RPST, the representational validity of real world problems—suggested as criterion for the RPG robustness by Barreteau et al. (2013) and Lennon et al. (2015)—is attained by the variety of the characters' profiles interpreted by teams of players in relation to the RPST ontology, related to the demonstrated trade-off mechanisms from the literature.

We were not able to test the influence of the players' background in real life (as public actors, private actors and civil society) with respect to the negotiation of the deals, type of arrangements, ES, and motivations. The 10 sessions were played by mixed-up heterogeneous groups of players with different backgrounds. It would have been appropriate to have multiple RPST sessions with homogeneous groups of players (all members of each of the 5 groups with the same profile of public actors, private actors and civil society) and to compare two or more sessions. Budget and time constraints limited us to one RPS session per location. Nevertheless, we did succeed in engaging players in reflecting on negotiation and on their own role in the ES flow, as an essential aspect of informing and setting ES governance and related arrangements.

5. Conclusions

This study proposed a comparative experimental approach based on role-playing simulation, applied in 10 case study areas in 8 Southeast European countries. The simulations proved that heterogeneous arrangements concerning different combinations of ES were proposed in the games to support ES delivery. We suggest that the variability of deals and ES entailed in the deals represent a group-based, rather than individual-based variability. In fact, the deals and related ES were not elicited by individuals (players) or appointed by characters (personified by teams of players) alone. Players' interactions generated novel insights and solutions for the delivered ES, based on a new shared understanding of trade-offs and synergies between ES.

Within decision-making processes, the emerging ES flow influenced by the arrangements on ES delivery was influenced by the local economic actor at community level, the Ecotour.

We verified that previous knowledge, such as ES mapping and assessments, can contribute to informed decisions. However, to activate and pass previous knowledge, communication and soft skills are essential. We demonstrated that ES flow emerges from the interplay of actors, and cannot be formulated or predicted only from previous studies of the ES supply capacity or of the ES demand potential. The deliberative process provides interactive mechanisms between players on ES trade-offs, resulting in different ES delivery and monetary exchange values. More research is needed with respect to (i) ES bundles emerging while negotiating arrangements to support the ES flow, with attention to the relationship between marketed and non-marketed ES, (ii) the influence of the players' background in the interaction process in decision making, (iii) the relative uptake of scientific and expert knowledge derived from ES assessment in real decision-making processes.

Role-playing games can be used to support learning-by-doing processes in decision-making for ES. Players can test the mechanisms of ES trade-off negotiation between them through direct personal interaction, besides the limitations of games representational reality.

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Appendix A

See Table A.1.

Appendix B

See Tables B.1–B.10 and Fig. B.1.

Table A.1
Geographic and social information of the locations where the RPST was played.

Id. Area	Country	Pilot area – landscape systems	Area (km^2)	RES*	Stakeholders and Target groups**	No. of participants to RPGS
1	AUSTRIA Eisendstad (AU)	Burgenland region, Eisenstadt, Gussing district	487	AS, FS, SE, HP, WP, RS (tot. 6)	Local Municipality, Technology Promotion Ltd., Centre for Renewable Energy Ltd., University, EU Funding Agency, Professional (energy service) Firm, Centre for Sustainable Rural Development, Regional and Local Development Agency, Provincial-Country Government, Mountain Community, Farmers, Non-governmental Organizations	33
2	BOSNIA HERZ. Republika Srpska (BH)	Banja Luka, Laktasi, Prnjavor and Gradiska area	388	AS, FS, SE, HP, WP, RS, GT (tot. 7)	Local Municipality, Association of Farmers, Agency for SME Development, Ministry of Agriculture, Forestry and Water Management, Association of Agronomists, Farmers, Local Innovation Centre, Public Health Institute, Non-governmental Organizations, Foreign Investments Promotion Agency, Grammar School, Electrotechnic school	18
3	BULGARIA Dimitrovgrad (BU)	Dimitrovgrad Municipality	62	AS, FS (Tot. 2)	Local Municipality, Technical and Information Support ltd, Civil registration Department of the local Municipality, Services ltd, Labour Office, Legal Department, Chemical Plant, Social Support ltd, Investment Control ltd, Energy company, Agricultural ltd, Economic activities ltd, Farmers, Private	22
4	ITALY 1 Rovigo (IT1)	Rovigo Province	1789	HP, GT (tot. 2)	Provincial-Country Government, Local Municipality, Association of Manufacturing, Association of Agronomists, Architectural Firm, University, Local development Agency, Environmental Association	14
5	ITALY 2 Verona (IT2)	Verona and its hills+	3109	–	Local Municipality, Regional Government, Association of Engineers, Architectural Firm, Agricultural Engineering (agronomist) Firm, University, Local Action Group, Institute of City Planning	13
6	KROATIA Istria Region (KR)	Cepic Field (Cepic Polje), Istria Region	~100	AS, FS, SE, WP, RS (tot. 5)	Local Municipality, University, Regional and Local Development Agency, Engineering Firm, Farmers, Environmental Association, Civil Groups, Students, Pensioner	28
7	ROMANIA Odorehiu (RO)	Municipality of Odorheiu Secuiesc	6639	AS, FS, SE, HP, WP (tot. 5)	Local Municipality, Farmers, Associations of Forest Owners, Agriculture Institute, Energy Company, Wood Processing Companies, Designers, Urban Services ltd, Non-governmental Organizations, Profession High school, Civil Groups	22
8	SLOVENIA Kranj (SL)	Gorenjska region (Kranj)	2137	SE, HP, WP, RS (tot. 4)	Local Energy agency, Public Forestry Service, Energy company, Agriculture and Forestry companies, Farmers, Students	9
9	HUNGARY 1 Szolnok (HU1)	Municipality of Szolnok	189	HP, GT (tot. 2)	Local Municipality, University, Non-governmental Organizations, Regional Energy Firm, Environmental and Energetic Firm, Environmental Energetic Engineers, Geologists, Planners, Agricultural workers, Farmers, Breeders, Retired People, Students, Citizens	29
10	HUNGARY 2 Ujszilvas (HU2)	Local Government of Ujszilvas	39	AS, SE, WP, GT (tot. 4)	Local Municipality, University, Non-governmental Organizations, Regional Institute of Social Support, Social Workers, Regional Energy Firm, Environmental and Energetic Firm, Environmental Energetic Engineers, Geologists, Planners, Agricultural workers, Farmers, Breeders, Entrepreneur, Retired People, Students, Citizens	35
	TOTAL					223

* AS = agricultural sector, FS = forestry sector, SE = solar energy, HP = hydro power, WP = wind power, RS = residues, GT = geothermal.
** Stakeholders and Target groups were identified by each local institution in its location, as reported in Gissi et al. (2014).
\+ Pilot area of the project "Rurbance", Alpine Space Programme – European Territorial Cooperation 2007–2013.

Table B.1
Deals emerging from the game sessions; TL = traditional landscape; IL = intensively cultivated landscape. They are classified in terms of types of arrangements (Arr*) and of types of transaction (Trans**); monetary exchanges are reported per each deal.

Play		Deals				ES	Monetary exchange	Revenues	Total
cod	id	Textual description	Synthesis	Arr*	Trans**	yes/no	Lak	Lak	Lak
RO	1	The Canka Province offers the possibility to elaborate ''green certificates'' for energy investments that uses renewable energy sources. In this way Sweet Energy Development Corporation receive 100 Lak	green certificates	M	DT	Y	100	–	100
	2	The Energy Corporation will buy the biomass produced by the farmers.	biomass feedstock	M	DT	Y	500	–	500
	3	Investments in environmental technological equipment	technological innovation	–	DT	N	200	–	200
	4	To increase the attractiveness of the natural reserve for the tourists and to improve tourists services	ecotourism services	–	DT	N	180	–	180
	5	Local tax for environmental protection to nourish a fund for habitat restoration	tax for environmental protection	H	T	Y	0	–	0
BH	6	The Canka Province will rent 100 km^2 of land (50 km^2 inside of TL and 50 km^2 outside of TL) to the Canka Farmers & Breeders Cooperative for 10 years with an obligation for Canka Cooperative to produce ecological yogurt while maintaining environmental characteristics of TL. During the first 3 years, the Canka Farmers & Breeders Cooperative will not pay taxes and will employ new workers. Later on will pay 100 Lak per year.	organic agriculture and food products	CH	TR	Y	0	−100	−100
	7	The Canka Province will rent 20 km^2 of land to Sweet Energy Development Corporation (ERP) that is included in the land used by the Canka Farmers & Breeders Cooperative. In the first 3 years, the Canka Province will not receive taxes and the plant for biomass has to be finished by ERP in 3 years.	biomass power plant	–	TR	N	0	−200	−200
	8	Sweet Energy Development Corporation (ERP) will provide Fund for environmental protection and ecotourism development in TL. ERP will pay to EcoTour in advance 50Lak and in the first three years 50 Lak. In the years following, ERP would pay 3% of its profit every year.	fund for environmental protection	CM	DT	Y	50	–	50
	9	Sweet Energy Development Corporation (ERP) will provide 200Lak to Canka Farmers & Breeders Cooperative as an investment and in return, Canka Farmers & Breeders Cooperative will deliver biomass to ERP.	biomass feedstock	CM	DT	Y	200	–	200
	10	EcoTour pays to Canka Farmers & Breeders Cooperative 25Lak immediately and 5% of profit per year to provide purification of wastewater with the conditions that they will not further expand in the following 10 years, and provide support and cooperation on expansion of tourism in the region.	Payments for ES	CM	DT	Y	25	–	25
SL	11	Canka Province Government will allow that 200 km^2 of TL is used for natural treatment of wastewater from members' farms, also for soil and flood prevention. This will allow Cooperative to keep on with production and paying 100 Lak of taxes. Cooperative will also pay to Government 150 Lak as they will save money from ''classic'' wastewater treatment planet.	Payments for ES	CH	DT	Y	150	–	150
	12	Canka Province Government will allow that 200 km^2 of TL is used for natural treatment of wastewater from farming. As this land will not be used for farming or other activities, they will be also available as living space for Red-footet falcon and eco-tours on minimum area of 200 km^2. This will bring to the Ecotour annual profit of 180 Lak, which is enough to pay 50 Lak of taxes to Government and still have 80 (130) Lak for operating.	ES maintenance	CH	R	Y	0	180	180
	13	Corporation will buy each year all biomass needed for operating of Energy Power planet, mainly from 80 km^2 of land, which will be affected by operating of the plant, for which it will plus additional 100 Lak per year. This will give Cooperative additional income from production and financial resources to pay annually 100 Lak of taxes.	biomass feedstock	M	DT	Y	100	–	100
	14	Canka Farmers&Breeders Cooperative will lease 20 km^2 of TL for 100 Lak to Sweet Energy Development	land leasing for biomass power plant	–	DT	N	100	–	100
	15	Canka Province Government will allow that 200 km^2 of TL is used for natural treatment of wastewater from members' farms, which will save to Cooperative money from ''classic'' wastewater treatment plant. Besides that, it will be possible to use this landscape for touristic tours of Ecotour. In order to ensure and monitor the ES provisioning, and also to ensure these landscapes are in good condition as a habitat of red footed falcon, Cooperative and Ecotour will each pay every year for these services 25 Lak to LandscapeAdvocates.org (LA), together 50 Lak. This is also the amount LA needs for its activities.	Monitoring ES; payments for ES	CH	DT	Y	50	0	50
HR	16	Canka Province will lease (20 years) to SweetEnergy 20 km^2 of land out of TL for building the plant and 80 km^2 in the TL (additional 20 km^2 will be leased from Farmers, so SweetEnergy will take care of 100 km^2 in TL that will be affected by the plant). SweetEnergy will pay 250 Lak to Province for leasing the land. Sweet Energy agrees to pay 200 taxes, and guarantees that will create new jobs and build infrastructure (roads, power grid) from which all residents of the area will have benefits. Sweet Energy agrees to comply all environmental regulations related to wastewater discharge, nutrients load and soil protection.	land leasing for biomass power plant	–	DT	N	250	–	250
	17	Canka Farmers & Breeders Cooperative agrees to build the water treatment plant and will pay to Canka Province 500 Lak, so Farm will comply with new law and can carry on with existing activities.	waste water treatment	–	DT	N	0	–	0
	18	Guarantee that at least 220 km^2 of TL will be untouched.	warranty for environmental protection (no new activities)	–	DT	N	25	–	25

	19	In cooperation with LandscapeAdvocates.org, Province will organize seminars for local stakeholder and target groups. Seminars will educate participant about the relevance of Ecosystem Services	capacity building – life-long learning on ES	–	DT	N	25	–	25
	20	Canka Province will ensure that Sweet energy restores (waste water and soil pollution) about 100 km^2 of TL that will be affected with new plant	Compensation for environmental restoration	H	A	Y	0	0	0
	21	EcoTours will pay 100 Lak to Province (from planned revenues of touristic activities) for conserving 200 km^2 of TL. EcoTour will pay 50 Lak of tax at end of this year.	environmental conservation for ES provisioning	CH	DT	Y	100	0	100
	22	SweetEnergy will pay 100 Lak to Canka Farmers and Breeders Cooperative for leasing 20 km^2 of TL (additional 80 km^2 will be leased from Province.	land leasing for biomass power plant	–	DT	N	100	–	100
	23	Sweet Energy guarantees that additional contracts will be made with local farmers, about buying the agricultural biomass for the plant. Farmers will have additional income from selling the biomass.	biomass feedstock	M	DT	Y	nd	–	nd
	24	LanscapeAdvocates agree to help Canka Farmers & Breeders Cooperative in building the water treatment plant and will give 50 Lak to help financing the plant. Plant will assure that Farmers do not pollute the environment, water and soil with their wastewater.	financing waste-water plant construction for environmental protection	–	DT	N	50	0	50
	25	As EcoTour has knowledge that conservation of 220 km^2 will bring more tourists (300 Lak of revenues for EcoTour), 50 Lak a year will be given to Farmers and they must invest it in development of rural tourism and tourists services maintaining local traditional organic food production.	Payments for ES	CM	DT	Y	50	250	300
	26	LandscapeAdvocates.org will lobby to provincial government to preserve at least 200 km^2 of TL. LandscapeAdvocates will give 50 Lak to EcoTour, and that money will be used for financing the conservation of TL. EcoTour will use that money for leasing the land from Farmers.	Lobbing for ES conservation	–	DT	N	50	–	50
HU1	27	Canka Farmers & Breeders Cooperative will build a wastewater treatment system. It costs 500 Lak. The Canka Provincial Government provides tax relief 50 Lak to the Canka Farmers & Breeders Cooperative to support its investments. Canka Farmers & Breeders Cooperative will build a new wastewater treatment plant with electric turbine. They can offer new workplace for local population and clean water and electricity free of charge to the Canka Province.	tax relief	–	TR	N	0	−50	−50
	28	The Canka Province rents out 20 km^2 land to Sweet Energy Development Corporation to build a biomass power plant. The Canka Province also needs to minimize risks of flood, or soil drainage problems, therefore Sweet Energy build a flood protection dam around the risky area. The Canka Province wants to increase employment, therefore supports Sweet Energy's investment. Sweet Energy will pay 150 Lak tax to start investment and 50 Lak for 20 km^2 and 10 Lak tax/year.	tax relief	–	DT, TR	N	50	−50	0
	29	The Canka Provincial Government rents 200 km^2 of TL to the Eco-Tours and Landscape Advocates for ecotourism activities, in order to preserve the TL ecological values. The Canka Provincial Government will receive 20 Lak tax per year. (Landscape Advocates will pay 80 Lak)	Supporting ES provisioning	CH	DT	Y	100	–	100
	30	Sweet Energy Development Corporation will cooperate with Canka Farmers & Breeders Cooperative. Sweet Energy Development Corporation will build a biomass plant and rent free of charge 19 km^2 to Canka Farmers & Breeders Cooperative. Canka Farmers & Breeders Cooperative will grow cereals as a biomass feedstock. During the construction, Canka Farmers & Breeders Cooperative will do the groundwork and provide biomass feedstock for 50 Lak/year.	biomass feedstock	M	DT	Y	50	–	50
	31	Canka Farmers & Breeders Cooperative rents 100 km^2 for 100 Lak to Eco-Tours. Canka Farmers & Breeders Cooperative will preserve TL needed to the Red-footed Falcon and TL ecological values.	Payment for ES	CM	DT	Y	100	–	100
	32	This is a lobby project together with Eco-Tours. The main aim to protect, enhance and conserve 200 km^2 territories as a TL. With Eco-Tours together asks Canka Province or/and Canka Cooperative to rent out 200 km^2 to provide good environment for Red-footed Falcon. Landscape Advocates has 80 Lak and Eco-Tours has 20 Lak to fix and rent 200 km^2 area. If they have incomes from 200 km^2 as tourism, etc., they would share them between each other.	Agreement for ES	CM	A	Y	0	–	0
HU2	33	The Canka Province provides 20 km^2 land for Sweet Energy to build a biomass plant. Sweet Energy will pay 120 Lak tax/year and provide free energy for public buildings, it costs 60 Lak.	power plant and tax relief	–	TR	N	0	−80	−80
	34	The Canka Province rents out 50 km^2 of surface to the Canka Farmers & Breeders Cooperative for 10 years for traditional low-impact farming for 50 Lak. It is forbidden to use this surface in a different way.	ES provisioning support	CH	S	Y	50	–	50
	35	The Canka Province will receive 72.50 Lak tax per year. Canka Farmers & Breeders Cooperative will build a wastewater treatment plant. It costs 500 Lak.	tax relief	–	TR	N	0	−28	−28
	36	The Canka Province rents 200 km^2 of surface to the Eco-Tours for 10 years for touristic activities. The Canka Province will receive 20 Lak tax per year.	tax relief/ES provisioning support	CH	TR	Y	0	−30	−30
	37	The Canka Province rents 100 km^2 of surface to the Landscape Advocates for 50 Lak to maintain the TL and its values	environmental conservation	H	DT	Y	50	0	50
	38	Sweet Energy Development Corporation will not build biomass plant in 400 km^2 TL territory. They will build it outside the TL area. Sweet Energy Development Corporation wants to get public support from environmentalists to the new biomass plant. Same deal signed with Eco-Tours to get support also. As deal shows, it has high importance for Sweet Energy Development Corporation, but low for Landscape Advocates.	Agreement for corporate environmental responsibility	–	A	N	0	0	0

(*continued on next page*)

Table B.1 (*continued*)

Play		Deals				ES	Monetary exchange	Revenues	Total
cod	id	Textual description	Synthesis	Arr*	Trans**	yes/no	Lak	Lak	Lak
	39	Sweet Energy Development Corporation will support Canka Farmers & Breeders Cooperative with 150 Lak to build a wastewater treatment. Sweet Energy provides free electricity and fuel to Canka Cooperative. Canka Cooperative offered biomass feedstock on barter. Together they will build flood protection dam, which gives possibility to produce more biomass feedstock.	waste water treatment/biomass feedstock	M	DT	Y	150	–	150
	40	Canka Farmers & Breeders Cooperative receives 50 Lak from Eco-Tours to preserve 100 km^2 of TL for Red-footed for touristic activities	Payment for ES	CM	DT	Y	50	–	50
	41	Eco-Tours agrees with Landscape advocates to form a coalition to protect, enhance and conserve healthy and productive TL. Joint touristic programs. Start a lobby activity for 300 km^2 to provide safe and secure environment for Red-footed Falcon.	Agreement for ES provisioning	–	A	N	0	0	0
	42	The Canka Farmers & Breeders Cooperative agrees with Landscape Advocated to preserve 100 km^2 of TL for traditional farming activities. For this Canka Farmers & Breeders Cooperative will receive 50 Lak per year.	Support ES provisioning	CM	DT	Y	50	–	50
AU	43	Establishment of a natural protected area in TL	environmental conservation	H	DT	Y	50	–	50
	44	Traditional cultivation of 80 km^2 and postponement of tax payment	tax relief/payments for ES	CH	DT, TR	Y	0	−100	−100
	45	20 km^2 of land to build the power plant, Sweet Energy builds flooding protection	power plant and flooding protection infrastructures	–	DT	N	0	0	0
	46	LA supports with a payment of 100 Lak sustainable agriculture in TL for 70 km^2, for technological improvement and innovative sustainable management of farms	Payments for ES	CM	DT	Y	100	–	100
	47	Sweet Energy buys feedstock from Farmers for 150 Lak	biomass feedstock	M	DT	Y	150	–	150
	48	Support in communication and raising awareness from LA to ET about TL ecological values, for 20 Lak	Communication Raising awareness	C	DT	Y	20	–	20
IT1	49	Lease of 20 km^2 in TL, lease of 100 km^2 in IL for feedstock production, protection of 50 km^2 in IL in compensation, concession of 50% of thermic energy for free	land lease for feedstock production-energy exchange	–	DT	N	150	–	150
	50	Bartle exchange of goods and services: employment for farmers; agricultural residuals as power plant feedstock; energy provision to farmers at lower costs	BBES exchange, employment and price reduction	M	A	Y	0	0	0
	51	Lobby for Power plant localization outside the TL; support in ecotourism activities as didactic farmers etc.	Lobbing to support ES provision	C	A	Y	0	–	0
	52	conversion of 100 km^2 of IL into TL for phytodepuration and other ecological benefits, between Canka Province and Farmers.	savings from not building the waste water treatment	CH	SA	Y	0	−500	−500
	53	Canka Province Gov. preserves 220 km^2in TL from, with a payment of 130 Lak from LA	ES provisioning support	CH	DT	Y	130	–	130
	54	Localization of the power plant out of the TL; buffer zone around the TL; non-invasive structures and compensation actions as territorial marketing, ecotourism infrastructures ad structures, support to ecotourism activities	Agreement for ES	CM	A	Y	0	0	0
	55	ecological restoration of 150 km^2 of TL by the Canka Province, voluntary doubling of taxes from Ecotour	Payments for ES	CH	DT	Y	50	–	50
IT2	56	Lease of 20 km^2 for power plant; compensation of 2 km^2, and payment of 25 Lak	land lease for power plant	–	DT	N	25	–	25
	57	Farmers lease 10 km^2 of TL to Sweet Energy, compensation for agricultural activities, concession of agricultural residuals in exchange with energy for agricultural activities	land lease for power plant, biomass feedstock/price reduction exchange	M	DT	Y	100	–	100
	58	Permit issuing in IL for the power plant construction, in exchange of the promotion of ES provisioning capacity of TL	ES provisioning support	–	A	N	0	0	0
	59	Agricultural buildings restoration for ecotourism activities	Ecotourism infrastructures and services	C	DT	Y	150	0	150
	60	Exploratory analysis for ecotourism eco-labelling, in exchange of organic agriculture and total natural reserve in 100 km^2 of TL	Eco-labelling (preliminary analysis)	C	A	Y	0	0	0
	61	Environmental reserve of 200 km^2 by Province; Analysis of biodiversity and important species from LA, soil and water systems; In case of biomass production in TL, government will require compensation; activity of communication and raising awareness with local communities	Agreement for ES	CH	A	Y	0	–	0
	62	Reduction of taxation of 50% for ecotourism, compensated from new income for the Province from forecasted tourism increase	tax relief	–	TR	N	0	−25	−25
	63	Organization of guided tours in biomass power plant; the plant will consume only 12 km^2 in TL by adopting new technologies; share of income from guided tours between the two.	Collaboration about RES activities	–	A	N	0	–	0

	64	Maintenance of 100 km^2 of TL for rural ecotourism through organic agriculture and agricultural activities; estimated income of 90 Lak will be shared between Ecotour and Farmers	Collaboration on ES providing activities	CM	R	Y	0	90	90
BU	65	Subsides to farmers from Provincial government to support TL ecological conservation through traditional agriculture	Subsidies	H	S	Y	100	0	100
	66	Biomass production from Farmers is sold to Energy Company	biomass feedstock	M	DT	Y	250	–	250
	67	Lease of 20 km^2 for power plant; compensation in 10 km^2 for generated impacts	land lease for power plant	–	DT	N	100	–	100

* Type of arrangements; C = Community-based arrangements, H = hierarchical approach, M = Market-driven approach, CH = hybrid of community-based and hierarchical approaches, CM = hybrid of community-based and market-driven approaches, HM = hybrid of hierarchical and market-driven approaches.

** Type of transaction; DT = direct transaction, as monetary exchange between characters; TR = Tax relief, suspension of tax payment by the Canka Province Government; S = subsidies; T = tax, imposition of a new tax for environmental conservation; A = agreement, any sort of agreement of commitments or exchange of services between partners that does not involve any money exchange; SA = savings by characters, e.g. for not building infrastructures; R = revenues from economic activities.

Table B.2
Distribution of deals per types of transaction in the 10 game sessions; deals are divided in the ones that include (with ES) or not include (without ES) ES negotiation. Type of transactions are: A = agreement, any sort of agreement of commitments or exchange of services between partners that does not involve any money exchange; DT = direct transaction, as monetary exchange between characters; M = multiple types among the ones mentioned; R = revenues from economic activities; S = subsidies; SA = savings by characters, e.g. for not building infrastructures; T = tax, imposition of a new tax for environmental conservation; TR = Tax relief, suspension of tax payment by the Canka Province Government.

		No. of deals per types of transactions										
Game sessions	Deals with or without ES	A	DT	M	R	S	SA	T	TR	Deals without ES	Deals with ES	Total number of deals
AU	without ES		1							1		6
	with ES		4	1							5	
BH	without ES								1	1		5
	with ES		3						1		4	
BU	without ES		1							1		3
	with ES		1			1					2	
HR	without ES		7							7		11
	with ES	1	3								4	
HU1	without ES			1					1	2		6
	with ES	1	3								4	
HU2	without ES	2							2	4		10
	with ES		4			1			1		6	
IT1	without ES		1							1		7
	with ES	3	2				1				6	
IT2	without ES	2	1						1	4		9
	with ES	2	2		1						5	
RO	without ES		2							2		5
	with ES		2					1			3	
SL	without ES		1							1		5
	with ES		3		1						4	
	tot	11	41	2	2	2	1	1	7	24	43	67

Table B.3
Spearman's correlation analysis between the types of arrangements in the deals and the RPST sessions, the Ecosystem Services (ES), the characters (CH) negotiating the deals, the motivations (M), and the players' background; values report moderate correlation (ρ = 0.40–0.49), strong correlations (ρ = 0.50–0.69), very strong correlations (ρ > 0.70).

Variables	Arrangement types
Game sessions	−0.054
Ecosystem services	
ES1 Water purification	0.322(**)
ES2 Flood prevention	0.263(*)
ES3 Habitat services	0.245(*)
ES4 Food production	0.198
ES5 BBES	−0.374(**)
ES6 Aesthetic values	0.373(**)
ES7 Recreational values	0.414(**)
ES8 Soil preservation	0.307(*)
Characters negotiating deals	
CH1. Canka Province government	−0.211
CH2. Sweet Energy Development Company	−0.327(**)
CH3. Farmers and Breeders Cooperative	−0.021
CH4. Landscape Advocates	0.211
CH5. Ecotour	0.401(**)
Motivations	
RPST-driven motivations (M1, M2, M3)	−0.389(**)
Motivations not driven by the RPST (M4, M5, M6)	0.209
M1. Economic (income/revenues)	−0.247(*)
M2. Legal	−0.223
M3. Economic (employment)	−0.312(*)
M4. Environmental	0.251(*)
M5. Public image	−0.049
M6. Values and moral	0.301(*)
Players' background	
Public actors	−0.086
Private actors	−0.085
Civil society	0.054

(*) Correlation is significant at the 0.05 level (2-tailed).
(**) Correlation is significant at the 0.01 level (2-tailed).

Table B.4
ES per deal in the 10 sessions.

Play cod	Deals id	ES1. Water purification p/a	ES2. Soil preservation p/a	ES3. Habitat services p/a	ES4. Food production p/a	ES5. BBES p/a	ES6. Aesthetic values p/a	ES7. Recreational values p/a	ES8. Flood prevention p/a
RO	1	–	–	–	–	1	–	–	–
	2	–	–	–	–	1	–	–	–
	3	–	–	–	–	–	–	–	–
	4	–	–	–	–	–	–	–	–
	5	–	–	1	–	–	–	1	–
BH	6	–	–	–	1	–	–	–	–
	7	–	–	–	–	–	–	–	–
	8	–	–	–	–	–	–	1	–
	9	–	–	–	–	1	–	–	–
	10	1	–	–	–	–	–	1	1
SL	11	1	–	–	–	–	–	–	–
	12	–	–	–	–	–	–	1	–
	13	–	–	–	–	1	–	–	–
	14	–	–	–	–	–	–	–	–
	15	–	–	1	–	–	–	–	–
HR	16	–	–	–	–	–	–	–	–
	17	–	–	–	–	–	–	–	–
	18	–	–	–	–	–	–	–	–
	19	–	–	–	–	–	–	–	–
	20	1	1	1	–	–	–	–	1
	21	1	1	1	–	–	1	1	1
	22	–	–	–	–	–	–	–	–
	23	–	–	–	–	1	–	–	–
	24	–	–	–	–	–	–	–	–

Table B.4 (*continued*)

Play cod	Deals id	ES1. Water purification p/a	ES2. Soil preservation p/a	ES3. Habitat services p/a	ES4. Food production p/a	ES5. BBES p/a	ES6. Aesthetic values p/a	ES7. Recreational values p/a	ES8. Flood prevention p/a
	25	–	–	–	1	–	–	1	–
	26	–	–	–	–	–	–	–	–
HU1	27	–	–	–	–	–	–	–	–
	28	–	–	–	–	–	–	–	–
	29	1	1	1	–	–	1	1	1
	30	–	–	–	–	1	–	–	–
	31	1	1	1	–	–	1	1	1
	32	1	1	1	–	–	1	1	1
HU2	33	–	–	–	–	–	–	–	–
	34	1	1	1	1	–	1	1	1
	35	–	–	–	–	–	–	–	–
	36	–	–	–	–	–	1	1	–
	37	1	1	1	–	–	1	1	1
	38	–	–	–	–	–	–	–	–
	39	–	–	–	–	1	–	–	–
	40	–	–	–	–	–	1	1	–
	41	–	–	–	–	–	–	–	–
	42	1	1	1	–	–	1	1	1
AU	43	–	–	–	–	–	1	1	–
	44	1	1	1	1	–	1	1	1
	45	–	–	–	–	–	–	–	–
	46	1	1	1	1	–	1	1	1
	47	–	–	–	–	1	–	–	–
	48	–	–	–	–	–	1	1	–
IT1	49	–	–	–	–	–	–	–	–
	50	–	–	–	–	1	–	–	–
	51	–	–	–	–	–	1	1	–
	52	1	1	–	–	–	–	–	1
	53	1	1	1	–	–	1	–	1
	54	–	–	–	–	–	1	1	–
	55	–	–	–	–	–	1	1	–
IT2	56	–	–	–	–	–	–	–	–
	57	–	–	–	–	1	–	–	–
	58	–	–	–	–	–	–	–	–
	59	–	–	–	–	–	–	1	–
	60	–	–	–	1	–	–	1	–
	61	1	1	1	–	–	1	1	1
	62	–	–	–	–	–	–	–	–
	63	–	–	–	–	–	–	–	–
	64	–	–	1	1	–	1	1	–
BU	65	–	–	–	1	–	–	–	–
	66	–	–	–	–	1	–	–	–
	67	–	–	–	–	–	–	–	–

Table B.5
Description of the ES negotiated in the deals of each game sessions; ES classes are mentioned several times in the deals of each game session.

	Number of deals mentioning ES class per game session											Sessions negotiating the ES class
Ecosystem services class	AU no.	BH no.	BU no.	HR no.	HU1 no.	HU2 no.	IT1 no.	IT2 no.	RO no.	SL no.	Tot no.	no.
ES1. Water purification	2	1	0	2	3	3	2	1	0	1	15	8
ES2. Soil preservation	2	0	0	2	3	3	2	1	0	0	13	6
ES3. Habitat services	2	0	0	2	3	3	1	2	1	1	15	8
ES4. Food production	2	1	1	1	0	1	0	2	0	0	8	6
ES5. BBES	1	1	1	1	1	1	1	1	2	1	11	10
ES6. Aesthetic values	4	0	0	1	3	5	4	2	0	0	19	6
ES7. Recreational values	4	2	0	2	3	5	3	4	1	1	25	9
ES8. Flood prevention	2	1	0	2	3	3	2	1	0	0	14	7
Total number of ES negotiated per game session	19	6	2	13	19	24	15	14	4	4	120	

Table B.6
Pearson's correlation between ES classes involved in the 43 deals; in bold the statistically significant values.

Pearson Correlation	ES1	ES2	ES3	ES4	ES5	ES6	ES7	ES8
Game sessions	0.017	0.11	0.022	0.171	−0.074	0.278(*)	0.125	0.057
Ecosystem services								
ES1 Water purification	1							
ES2 Flood prevention	**0.914(**)**	1						
ES3 Habitat services	**0.742(**)**	**0.823(**)**	1					
ES4 Food production	0.133	0.169	0.244(*)	1				
ES5 BBES	−0.238	−0.217	−0.238	−0.163	1			
ES6 Aesthetic values	0.536(**)	0.612(**)	0.615(**)	0.177	−0.279(*)	1		
ES7 Recreational values	0.400(**)	0.402(**)	0.474(**)	0.287(*)	−0.342(**)	**0.747(**)**	1	
ES8 Soil preservation	**0.957(**)**	**0.955(**)**	**0.781(**)**	0.150	−0.228	0.573(**)	0.438(**)	1

(*) Correlation is significant at the 0.05 level (2-tailed).
(**) Correlation is significant at the 0.01 level (2-tailed).

Table B.7
Motivations per play.

	AU		BH		BU		IT1		IT2		HR		RO		SL		HU1		HU2		Total	
Motivations	no.	%	no.	%	no.	%	no.	%	no.	%	no.	%	no.	%	no.	%	no.	%	no.	%	no.	%
M1 economic	5	17.2	5	22.7	3	33.3	6	27.3	8	29.6	11	25.0	5	29.4	5	19.2	6	30.0	8	22.9	62	24.7
M2 legal	4	13.8	2	9.1	1	11.1	1	4.5	2	7.4	3	6.8	3	17.6	3	11.5	0	0.0	4	11.4	23	9.2
M3 employment	5	17.2	4	18.2	3	33.3	3	13.6	4	14.8	6	13.6	2	11.8	3	11.5	5	25.0	4	11.4	39	15.5
M4 environmental	5	17.2	4	18.2	2	22.2	6	27.3	8	29.6	8	18.2	3	17.6	5	19.2	6	30.0	10	28.6	57	22.7
M5 public image	4	13.8	5	22.7	0	0.0	3	13.6	3	11.1	8	18.2	2	11.8	5	19.2	2	10.0	2	5.7	34	13.5
M6 values & moral	6	20.7	2	9.1	0	0.0	3	13.6	2	7.4	8	18.2	2	11.8	5	19.2	1	5.0	7	20.0	36	14.3
total	29	100	22	100	9	100	22	100	27	100	44	100	17	100	26	100	20	100	35	100	251	100
ES – total	19		6		2		15		14		13		4		4		19		24		120	
Deals - total	6		5		3		7		9		11		5		5		6		10		67	
ES per deal	3.17		1.2		0.67		2.1		1.6		1.2		0.8		0.8		3		2.4		1.8	
Motivations per deal	4.8		4.4		3.0		3.1		3.0		4.0		3.4		5.2		3.3		3.5		3.7	
Motivations per ES	1.5		3.7		4.5		1.5		1.9		3.4		4.3		6.5		1.1		1.5		2.1	

Table B.8
Spearman's correlation analysis between motivations and game sessions, ecosystem services (ES), characters (CH) negotiating deals and players' backgrounds; M1 = economic (income/revenues), M2 = legal, M3 = economic (employment), M4 = environmental, M5 = public image, M6 = values & moral.

Variables	M1	M2	M3	M4	M5	M6
Game sessions	−0.08	−0.214	−0.081	−0.098	−0.233(**)	−0.266(*)
Ecosystem services						
ES1. Water purification	−0.202	−0.015	−0.083	0.107	−0.011	0.12
ES2. Soil preservation	−0.197	−0.149	−0.056	0.062	−0.115	0.056
ES3. Habitat services	−0.039	−0.104	−0.13	0.232	0.006	0.083
ES4. Food production	−0.113	−0.067	0.118	−0.005	−0.05	0.009
ES5. BBES	0.326(**)	0.055	0.307(*)	−0.360(**)	−0.066	−0.211
ES6. Aesthetic values	−0.128	−0.097	−0.052	0.199	−0.151	0.009
ES7. Recreational values	−0.217	−0.069	−0.084	0.260(*)	0.042	0.092
ES8. Flood prevention	−0.234	−0.08	−0.046	0.085	−0.062	0.106
Characters negotiating deals						
CH1. Canka Province Government	−0.057	0.320(**)	0.027	−0.11	−0.118	−0.006
CH2. Sweet Energy Development Company	0.310(*)	0.104	0.345(**)	−0.335(**)	−0.117	−0.281(*)
CH3. Farmers and Breeders Cooperative	0.192	−0.016	0.084	−0.184	−0.064	−0.097
CH4. Landscape Advocates	−0.336(**)	−0.347(**)	−0.329(**)	0.499(**)	0.215	0.386(**)
CH5. Ecotour	−0.065	−0.157	−0.209	0.255(*)	0.1	0.062
Players' background						
Public actors	−0.251(*)	0.072	0.057	0.103	−0.106	0.134
Private actors	−0.136	0.069	−0.098	0.059	0.066	0.043
Civil society	−0.166	−0.251	−0.069	−0.139	−0.112	0.047

(*) Correlation is significant at the 0.05 level (2-tailed).
(**) Correlation is significant at the 0.01 level (2-tailed).

Table B.9
Regression models of ES classes and individual characters' motivations, with Stepwise method (Criteria: Probability-of-F-to-enter <= 0.050, Probability-of-F-to-remove >= 0.100).

Dependent Variable	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics					Model	Unstandardized Coefficients		Std. Coeff.	t	Sig.	Correlations	
					R Square Change	F Change	df1	df2	Sig. F Change		B	Std. Error	Beta			Part	SR*
ES1	0.593(a)	0.352	0.342	0.406	0.352	35.303	1	65	0.000	(Constant)	0.745	0.059		12.562	0.000		
										M2 Cha2	-0.645	0.109	−0.593	−5.942	0.000	−0.593	35%
ES2	0.572(a)	0.327	0.317	0.415	0.327	31.632	1	65	0.000	(Constant)	0.723	0.061		11.945	0.000		
										M2 Cha2	−0.623	0.111	−0.572	−5.624	0.000	−0.572	33%
ES3	0.615(a)	0.378	0.369	0.397	0.378	39.552	1	65	0.000	(Constant)	0.766	0.058		13.239	0.000		
										M2 Cha2	−0.666	0.106	−0.615	−6.289	0.000	−0.615	38%
ES4	0.380(a)	**0.145**	0.132	0.409	0.145	10.995	1	65	0.001	(Constant)	0.362	0.060		6.069	0.000		
										M2 Cha2	−0.362	0.109	−0.380	−3.316	0.001	−0.380	14%
ES5	0.546(a)	0.298	0.287	0.326	0.298	27.597	1	65	0.000	(Constant)	0.043	0.048		0.894	0.000		
										M2 Cha2	0.457	0.087	0.546	5.253	0.000	0.546	30%
ES6	0.661(a)	**0.437**	0.428	0.374	0.437	50.435	1	65	0.000	(Constant)	0.809	0.055		14.833	0.000		
										M2 Cha2	−0.709	0.100	−0.661	−7.102	0.000	−0.661	44%
ES7	0.661(a)	**0.437**	0.428	0.374	0.437	50.435	1	65	0.000	(Constant)	0.809	0.055		14.833	0.000		
										M2 Cha2	−0.709	0.100	−0.661	−7.102	0.000	−0.661	44%
ES8	0.593(a)	0.352	0.342	0.406	0.352	35.303	1	65	0.000	(Constant)	0.745	0.059		12.562	0.000		
										M2 Cha2	−0.645	0.109	−0.593	−5.942	0.000	−0.593	35%

* Coefficients of part correlation Square.

Table B.10
Ranges of monetary values negotiated in the deals involving ES classes; the monetary exchange values associated to the BBES provisioning services in was negotiated individually; in bold the values for BBES exchanges, the only provisioning service exchanged alone.

	Monetary values of deals involving ES classes				Negotiation between ES	
ES classes	mean Lak	min Lak	max Lak	range min–max Lak	Individually no.	With other ES no.
ES1 – Water purification	86	25	150	125	1	14
ES2 – Flood prevention	85	50	130	80	0	13
ES3 – Habitat services	81	50	130	80	1	14
ES4 – Food production	75	50	100	50	2	6
ES5 – BBES	**194**	**100**	**500**	**400**	**11**	**0**
ES6 – Aesthetic values	71	20	130	110	0	19
ES7 – Recreational values	66	20	150	130	3	22
ES8 – Soil preservation	78	25	130	105	0	14

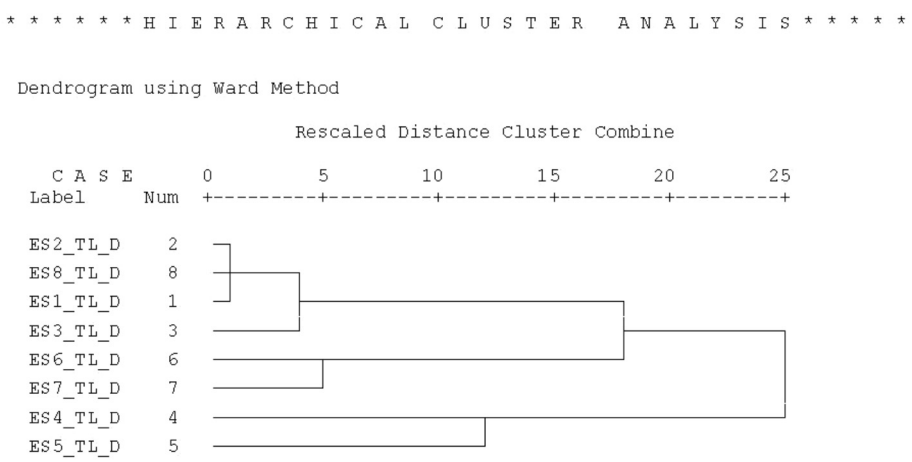


Fig. B.1. Hierarchical Cluster Analysis in the RPGS relating to Traditional Landscapes (TL).

Appendix C. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.ecoser.2018.03.025.

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